

ParTIpy - Methods

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1 Notation

- $N \in \mathbb{N}$ — number of samples (here: cells)
- $D \in \mathbb{N}$ — number of dimensions
- $G \in \mathbb{N}$ — number of genes
- $K \in \mathbb{N}$ with $K \leq N$ — number of archetypes
- $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ with $\mathbf{x}_n \in \mathbb{R}^D$ — embedded coordinates of all cells
- $\mathbf{X} \in \mathbb{R}^{N \times D}$ — data matrix whose n 'th row is $\mathbf{x}_n^T$
- $\mathcal{Z} = \{\mathbf{z}_1, \dots, \mathbf{z}_K\}$ with $\mathbf{z}_k \in \mathbb{R}^D$ — coordinates of the K archetypes
- $\mathbf{Z} \in \mathbb{R}^{K \times D}$ — archetype matrix whose k 'th row is $\mathbf{z}_k^T$
- $\mathbf{A} \in \mathbb{R}^{N \times K}$ — coefficients that define each sample as convex combinaton of archetypes
- $\mathbf{B} \in \mathbb{R}^{K \times N}$ — coefficients that define each archetype as convex combination of samples
- $\mathbf{Y}^{(\mathbf{X})} \in \mathbb{R}^{N \times G}$ — observed gene expression matrix; row n corresponds to cell n
- $\mathbf{Y}^{(\mathbf{Z})} \in \mathbb{R}^{K \times G}$ — inferred archetype gene expression matrix; row k corresponds to archetype k
- $\mathbf{1}_K \in \mathbb{R}^K$ — column vector of ones
- $\mathbf{I}_K \in \mathbb{R}^{K \times K}$ — $K \times K$ identity matrix
- $\mathbf{1}[\cdot]$ — indicator function

2 Archetypal Analysis

2.1 Objective

Let $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}_{n=1}^N$ be a data set comprising N D -dimensional data points, and let $\mathbf{X} \in \mathbb{R}^{N \times D}$ be the matrix where each row is a data point.

In Archetypal Analysis we make two assumptions:

1. Each data point is a convex combination of K archetypes;
2. Each archetype is a convex combination of N data points.

Expressing the first assumption in matrix notation yields

$$\hat{\mathbf{X}} = \mathbf{AZ} \quad \text{or} \quad \hat{\mathbf{x}}_n = \mathbf{Z}^T \mathbf{a}_n \quad \text{for } n = 1, \dots, N \quad (1)$$

where $\hat{\mathbf{X}} \in \mathbb{R}^{N \times D}$ is the reconstructed data matrix, $\mathbf{Z} \in \mathbb{R}^{K \times D}$ is the matrix of archetypes (i.e. each row is one archetype), and $\mathbf{A} \in \mathbb{R}^{N \times K}$ is a row-stochastic matrix that defines by which archetypes each data point is formed.

Expressing the second assumption in matrix notation yields

$$\mathbf{Z} = \mathbf{BX} \quad \text{or} \quad \mathbf{z}_k = \mathbf{X}^T \mathbf{b}_k \quad \text{for } k = 1, \dots, K \quad (2)$$

where $\mathbf{B} \in \mathbb{R}^{K \times N}$ is a row-stochastic matrix specifying the contribution of each data point to the construction of each archetype.

Here, we will quantify the reconstruction error using the RSS, given by the squared Frobenius norm,

$$\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 = \|\mathbf{X} - \mathbf{AZ}\|_F^2 = \|\mathbf{X} - \mathbf{ABX}\|_F^2 \quad (3)$$

which yields the following optimization objective

$$\begin{aligned} \hat{\mathbf{A}}, \hat{\mathbf{B}} = \arg \min_{\substack{\mathbf{A} \in \mathbb{R}^{N \times K} \\ \mathbf{B} \in \mathbb{R}^{K \times N}}} \|\mathbf{X} - \mathbf{ABX}\|_F^2 \quad \text{subject to} \\ \mathbf{A} \geq 0, \quad \mathbf{A}\mathbf{1}_K = \mathbf{1}_N \\ \mathbf{B} \geq 0, \quad \mathbf{B}\mathbf{1}_N = \mathbf{1}_K \end{aligned} \quad (4)$$

Introducing the set of row-stochastic matrices,

$$F(N, K) := \{\mathbf{A} \in \mathbb{R}^{N \times K} \mid \mathbf{A} \geq 0 \wedge \mathbf{A}\mathbf{1}_K = \mathbf{1}_N\} \quad (5)$$

we can write the objective compactly as:

$$\hat{\mathbf{A}}, \hat{\mathbf{B}} = \arg \min_{\substack{\mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N)}} \|\mathbf{X} - \mathbf{ABX}\|_F^2 \quad (6)$$

2.2 Properties of the Objective

2.2.1 Translation Invariance

The minimizers $\mathbf{A}, \mathbf{B}$ of the objective are invariant under row-wise translations of $\mathbf{X}$.

Let $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{1}_N \mathbf{v}^T$ for any $\mathbf{v} \in \mathbb{R}^D$, then

$$\arg \min_{\substack{\mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N)}} \|\tilde{\mathbf{X}} - \mathbf{A}\tilde{\mathbf{B}}\tilde{\mathbf{X}}\|_F^2 = \arg \min_{\substack{\mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N)}} \|\mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{X}\|_F^2 \quad (7)$$

For completeness, we provide our own derivation of this result in Section 6.1, following the approach of Mørup and Hansen (2012).

2.2.2 Scale Invariance

The minimizers $\mathbf{A}, \mathbf{B}$ of the objective are invariant under global scaling of $\mathbf{X}$. Let $\tilde{\mathbf{X}} = \lambda \mathbf{X}$ for any $\lambda \neq 0$, then

$$\arg \min_{\substack{\mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N)}} \|\tilde{\mathbf{X}} - \mathbf{A}\tilde{\mathbf{B}}\tilde{\mathbf{X}}\|_F^2 = \arg \min_{\substack{\mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N)}} \|\mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{X}\|_F^2 \quad (8)$$

For completeness, we provide our own derivation of this result in Section 6.2, following the approach of Mørup and Hansen (2012).

2.2.3 Uniqueness up to Permutation of Archetypes

Assuming that for each archetype k , there exists one data point n that is best reconstructed using only this archetype, i.e.

$$\forall k \in \{1, \dots, K\} \exists n \in \{1, \dots, N\} a_{nk} = 1 \quad (9)$$

and that for each archetype there exists one data point that is only used to define this archetype and not any other archetype, i.e.

$$\forall k \in \{1, \dots, K\} \exists n \in \{1, \dots, N\} b_{kn} > 0 \wedge b_{k'n} = 0 \forall k' \neq k \quad (10)$$

then the objective does not suffer from rotational ambiguity. I.e. any orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{K \times K}$ that satisfies

$$\begin{aligned} \|\mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{X}\|_F^2 &= \|\mathbf{X} - \mathbf{A}\mathbf{Q}\mathbf{Q}^{-1}\mathbf{B}\mathbf{X}\|_F^2 \\ &= \|\mathbf{X} - \tilde{\mathbf{A}}\tilde{\mathbf{B}}\mathbf{X}\|_F^2 \end{aligned} \quad (11)$$

must be a permutation matrix.

For completeness, we provide our own derivation of this result in Section 6.3, following the approach of Mørup and Hansen (2012).

2.2.4 Convexity of Subproblems

If we measure the reconstruction error with the RSS, the objective in Eq. (4) is biconvex: it is convex in $\mathbf{A}$ when $\mathbf{B}$ is fixed and vice-versa. We prove convexity in $\mathbf{A}$; the argument for $\mathbf{B}$ is analogous.

This biconvexity property is crucial for alternating optimization approaches, as it guarantees that each subproblem has a unique global optimum.

For completeness, we provide our own derivation of this result in Section 6.4, following the approach of Mørup and Hansen (2012).

2.2.5 Only Samples Outside of the Archetypal Convex Hull Contribute to the Loss

The objective in Equation 4 can be expressed as a sum over individual data points

$$\|\mathbf{X} - \mathbf{ABX}\|_F^2 = \sum_{n=1}^N \|\mathbf{x}_n - \mathbf{Z}^T \mathbf{a}_n\|_2^2. \quad (12)$$

For fixed archetypes $\mathbf{Z}$, the optimal coefficient matrix $\mathbf{A}$ assign each data point to its Euclidean projection onto the convex hull of the archetypes $\mathbf{z}_1, \dots, \mathbf{z}_K$, that is

$$\|\mathbf{X} - \mathbf{ABX}\|_F^2 = \sum_{n=1}^N \min_{\mathbf{q} \in \text{conv}(\mathcal{Z})} \|\mathbf{x}_n - \mathbf{q}\|_2^2 \quad (13)$$

Hence, any point $\mathbf{x}_n \in \text{conv}(\mathcal{Z})$ does not contribute to the loss.

2.3 Optimization

While the objective in Equation 4 is non-convex, it is biconvex, meaning that it becomes convex in $\mathbf{A}$ when $\mathbf{B}$ is fixed, and vice versa (see Section 2.2.4). A common strategy for optimizing such biconvex objectives is to initialize $\mathbf{B}$ (and thereby $\mathbf{Z}$) and then alternate between solving the convex subproblem in one variable while keeping the other fixed. This alternating minimization procedure results in Algorithm 1.

Algorithm 1 Prototypical Algorithm

- 1: **Input:** Data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$, number of archetypes K , max iterations T
- 2: **Initialize:** Archetypes $\mathbf{Z} \in \mathbb{R}^{K \times D}$
- 3: **for** $t = 1$ to T **do**
- 4: Compute optimal weights $\mathbf{A} \in \mathbb{R}^{N \times K}$:

$$\mathbf{A} \leftarrow \arg \min_{\mathbf{A} \in F(N, K)} \|\mathbf{X} - \mathbf{AZ}\|_F^2 \quad (14)$$

- 5: Compute optimal weights $\mathbf{B} \in \mathbb{R}^{K \times N}$:

$$\mathbf{B} \leftarrow \arg \min_{\mathbf{B} \in F(K, N)} \|\mathbf{X} - \mathbf{ABX}\|_F^2 \quad (15)$$

- 6: Update archetypes:

$$\mathbf{Z} \leftarrow \mathbf{BX} \quad (16)$$

- 7: **if** convergence criterion is met **then**
 - 8: **break**
 - 9: **end if**
 - 10: **end for**
 - 11: **Return:** $\mathbf{A}, \mathbf{B}, \mathbf{Z}$
-

Building on the alternating scheme in Algorithm 1, we implement three solver variants—(i) regularised non-negative least squares (RNNLS) (Cutler and Breiman 1994), (ii) principal convex-hull analysis (PCHA) (Mørup and Hansen 2012), and (iii) a projection-free Frank-Wolfe (FW) update (Bauckhage et al. 2015) - each differing in how the simplex constraints are enforced. The next three subsections detail these variants. For a broader survey of alternative optimisers, the reader is referred to Alcacer et al. (2025).

2.3.1 Regularized Nonnegative Least Squares (RNNLS)

Introduced by Cutler and Breiman (1994), this was the first algorithm to solve the archetypal analysis objective in Equation (4).

The authors proposed to solve the constrained optimization problems in lines 4 and 5 of Algorithm 1 using a Nonnegative Least Squares (NNLS) solver and enforcing the convexity constraints using a penalty term with regularization parameter λ . So for each sample $\mathbf{a}_1, \dots, \mathbf{a}_N$ (i.e. each row in $\mathbf{A}$) we solve

$$\begin{aligned} \mathbf{a}_n &= \arg \min_{\mathbf{a}_n \in \mathbb{R}^K} \|\mathbf{x}_n - \mathbf{Z}^T \mathbf{a}_n\|_2^2 + \lambda (1 - \mathbf{1}_K^T \mathbf{a}_n)^2 \quad \text{subject to} \quad \mathbf{a}_n \geq 0 \\ &\approx \arg \min_{\mathbf{a}_n \in \mathbb{R}^K} \left\| \begin{bmatrix} \mathbf{x}_n \\ \lambda \end{bmatrix} - \begin{bmatrix} \mathbf{Z}^T \\ \lambda \mathbf{1}_K^T \end{bmatrix} \mathbf{a}_n \right\|_2^2 \quad \text{subject to} \quad \mathbf{a}_n \geq 0 \end{aligned} \quad (17)$$

Then to optimize $\mathbf{B}$, we first compute the optimal $\mathbf{Z}$ given the current $\mathbf{A}$ using a standard least squares solver, for example the one implemented in `numpy`.

$$\mathbf{Z} = \arg \min_{\mathbf{Z} \in \mathbb{R}^{K \times D}} \|\mathbf{X} - \mathbf{A}\mathbf{Z}\|_F^2. \quad (18)$$

Then for each archetype $\mathbf{b}_1, \dots, \mathbf{b}_K$ (i.e. each row in $\mathbf{B}$) we solve

$$\begin{aligned} \mathbf{b}_k &= \arg \min_{\mathbf{b}_k \in \mathbb{R}^N} \|\mathbf{z}_k - \mathbf{X}^T \mathbf{b}_k\|_2^2 + \lambda (1 - \mathbf{1}_N^T \mathbf{b}_k)^2 \quad \text{subject to} \quad \mathbf{b}_k \geq 0 \\ &\approx \arg \min_{\mathbf{b}_k \in \mathbb{R}^N} \left\| \begin{bmatrix} \mathbf{z}_k \\ \lambda \end{bmatrix} - \begin{bmatrix} \mathbf{X}^T \\ \lambda \mathbf{1}_N^T \end{bmatrix} \mathbf{b}_k \right\|_2^2 \quad \text{subject to} \quad \mathbf{b}_k \geq 0 \end{aligned} \quad (19)$$

2.3.2 Principal Convex Hull Analysis (PCHA)

Inspired by the projected gradient method for NMF (Lin 2007) and normalization invariance approach introduced for NMF (Eggert and Korner 2004), the principal convex hull analysis (PCHA) algorithm was introduced by (Mørup and Hansen 2012) to solve the archetypal analysis objective in Equation (4).

First, we recast the optimization problem in terms of the ℓ_1 -normalization invariant variables $\tilde{a}_n$ and $\tilde{b}_k$ (called invariant because these variables won't change if one applies ℓ_1 -normalization).

$$\tilde{a}_{nk} = \frac{\max(a_{nk}, 0)}{\sum_{k''=1}^K \max(a_{nk''}, 0)}, \quad \tilde{b}_{kn} = \frac{\max(b_{kn}, 0)}{\sum_{n''=1}^N \max(b_{kn''}, 0)} \quad (20)$$

Thus our objective becomes

$$\hat{\mathbf{A}}, \hat{\mathbf{B}} = \arg \min_{\substack{\mathbf{A} \in \mathbb{R}^{N \times K} \\ \mathbf{B} \in \mathbb{R}^{K \times N}}} \|\mathbf{X} - \tilde{\mathbf{A}}\tilde{\mathbf{B}}\mathbf{X}\|_F^2 \quad \text{with} \quad \tilde{\mathbf{A}} = P_{\Delta_{K-1}}(\mathbf{A}), \tilde{\mathbf{B}} = P_{\Delta_{N-1}}(\mathbf{B}) \quad (21)$$

where we define P_{Δ_M} as the rowwise projection onto the M simplex, i.e. for any matrix $\mathbf{H} \in \mathbb{R}^{N \times M}$ we have

$$[P_{\Delta_M}(\mathbf{H})]_{nm} = \frac{\max(\mathbf{H}_{nm}, 0)}{\sum_{m'=1}^M \max(\mathbf{H}_{nm'}, 0)} \quad (22)$$

Then the gradient of the RSS with respect to $\mathbf{a}_n$ is obtained via the chain rule

$$\frac{\partial \text{RSS}}{\partial \mathbf{a}_n} = \frac{\partial \text{RSS}}{\partial \tilde{\mathbf{a}}_n} \frac{\partial \tilde{\mathbf{a}}_n}{\partial \mathbf{a}_n} \quad (23)$$

The first part (see Section 6.7) is given by

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial \tilde{\mathbf{a}}_n} &= 2 (\tilde{\mathbf{a}}_n \mathbf{Z} \mathbf{Z}^T - \mathbf{x}_n \mathbf{Z}^T)^T \\ &= (\tilde{g}_n^{(\mathbf{A})})^T \end{aligned} \quad (24)$$

The second part (see Section 6.8) is given by

$$\frac{\partial \tilde{\mathbf{a}}_n}{\partial \mathbf{a}_n} = \left(\frac{\left(\sum_{k''=1}^K a_{nk''} \right) \mathbf{I}_K - a_n \mathbf{1}_K^T}{\left(\sum_{k''=1}^K a_{nk''} \right)^2} \right) \text{diag} [\mathbf{1} [a_1 \geq 0] \quad \dots \quad \mathbf{1} [a_K \geq 0]] \quad (25)$$

Together we have

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial \mathbf{a}_n} &= \frac{\partial \text{RSS}}{\partial \tilde{\mathbf{a}}_n} \frac{\partial \tilde{\mathbf{a}}_n}{\partial \mathbf{a}_n} \\ &= (\tilde{g}_n^{(\mathbf{A})})^T \left(\frac{\left(\sum_{k''=1}^K a_{nk''} \right) \mathbf{I}_K - a_n \mathbf{1}_K^T}{\left(\sum_{k''=1}^K a_{nk''} \right)^2} \right) \text{diag} [\mathbf{1} [a_1 \geq 0] \quad \dots \quad \mathbf{1} [a_K \geq 0]] \end{aligned} \quad (26)$$

If we assume that $\mathbf{a}_n$ has been ℓ_1 normalized in the previous iteration (which we can easily do, since ℓ_1 normalization does not change the objective) the derivative simplifies to

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial \mathbf{a}_n} &= (\tilde{g}_n^{(\mathbf{A})})^T (\mathbf{I}_K - a_n \mathbf{1}_K^T) \\ &= (\tilde{g}_n^{(A)})^T \mathbf{I}_K - (\tilde{g}_n^{(A)})^T a_n \mathbf{1}_K^T \\ &= (\tilde{g}_n^{(A)})^T - \left(\sum_{k''=1}^K \tilde{g}_{nk''}^{(A)} a_{nk''} \right) \mathbf{1}_K^T \end{aligned} \quad (27)$$

So for a single element we have

$$\frac{\partial \text{RSS}}{\partial a_{nk}} = \tilde{g}_{nk}^{(A)} - \sum_{k''=1}^K \tilde{g}_{nk''}^{(A)} a_{nk''}. \quad (28)$$

which is the same as in Section 2.2 of (Mørup and Hansen 2012).

The algorithm to update $\mathbf{A}$ is given in Algorithm 2. The gradient descent step sizes are determined using line search.

Algorithm 2 Update $\mathbf{A}$ via Principal Convex Hull Analysis (PCHA)

```

1: Input:  $\mathbf{X}, \tilde{\mathbf{A}}, \mathbf{Z}$ 
2: Output: Updated  $\mathbf{A}$ 
3:  $\text{RSS} \leftarrow \|\mathbf{X} - \mathbf{AZ}\|_F^2$ 
4:  $\mu \leftarrow 1$ 
5: for  $t = 1$  to  $T$  do
6:    $\tilde{\mathbf{G}}^{(\mathbf{A})} \leftarrow 2 \left( \tilde{\mathbf{A}}\mathbf{Z}\mathbf{Z}^T - \mathbf{X}\mathbf{Z}^T \right)$ 
7:    $\mathbf{G}^{(\mathbf{A})} \leftarrow \tilde{\mathbf{G}}^{(\mathbf{A})} - \left( \tilde{\mathbf{G}}^{(\mathbf{A})} \odot \tilde{\mathbf{A}} \right) \mathbf{1}_K \mathbf{1}_K^T$ 
8:   for  $t^{(\text{line})} = 1$  to  $T^{(\text{line})}$  do ▷ line search
9:      $\mathbf{A} \leftarrow \mathbf{A} - \mu \mathbf{G}^{(\mathbf{A})}$ 
10:     $\tilde{\mathbf{A}} \leftarrow P_{\Delta_{K-1}}(\mathbf{A})$ 
11:     $\text{RSS}^{(t)} \leftarrow \|\mathbf{X} - \tilde{\mathbf{A}}\mathbf{Z}\|_F^2$ 
12:    if  $\text{RSS}^{(t)} < (1 + \epsilon) \cdot \text{RSS}$  then
13:       $\mu \leftarrow 1.2 \cdot \mu$ 
14:       $\text{RSS} \leftarrow \text{RSS}^{(t)}$ 
15:      break
16:    else
17:       $\mu \leftarrow 0.5 \cdot \mu$ 
18:    end if
19:  end for
20: end for
21: Return:  $\tilde{\mathbf{A}}$ 

```

Similarly, under the assumption that $\mathbf{b}_k$ has been ℓ_1 normalized in the previous iteration, the gradient with respect to $\mathbf{b}_k$ is given by:

$$\frac{\partial \text{RSS}}{\partial b_k} = \left(\tilde{g}_k^{(B)} \right)^T \mathbf{I}_N - \left(\sum_{n''=1}^N \tilde{g}_{kn''}^{(B)} b_{kn''} \right) \mathbf{1}_N^T \quad (29)$$

The algorithm to update $\mathbf{B}$ is given in Algorithm 3.

Algorithm 3 Update $\mathbf{B}$ via Principal Convex Hull Analysis (PCHA)

```

1: Input:  $\mathbf{X}, \tilde{\mathbf{A}}, \tilde{\mathbf{B}}$ 
2: Output: Updated  $\mathbf{B}$ 
3:  $\text{RSS} \leftarrow \|\mathbf{X} - \tilde{\mathbf{A}}\tilde{\mathbf{B}}\mathbf{X}\|_F^2$ 
4:  $\mu \leftarrow 1$ 
5: for  $t = 1$  to  $T$  do
6:    $\tilde{\mathbf{G}}^{(\mathbf{B})} \leftarrow 2 \left( \tilde{\mathbf{A}}^T \tilde{\mathbf{A}} \tilde{\mathbf{B}} \mathbf{X} \mathbf{X}^T - \tilde{\mathbf{A}}^T \mathbf{X} \mathbf{X}^T \right)$ 
7:    $\mathbf{G}^{(\mathbf{B})} \leftarrow \tilde{\mathbf{G}}^{(\mathbf{B})} - \left( \tilde{\mathbf{G}}^{(\mathbf{B})} \odot \tilde{\mathbf{B}} \right) \mathbf{1}_N \mathbf{1}_N^T$ 
8:   for  $t^{(\text{line})} = 1$  to  $T^{(\text{line})}$  do ▷ line search
9:      $\mathbf{B} \leftarrow \mathbf{B} - \mu \mathbf{G}^{(\mathbf{B})}$ 
10:     $\tilde{\mathbf{B}} \leftarrow P_{\Delta_{N-1}}(\mathbf{B})$ 
11:     $\text{RSS}^{(t)} \leftarrow \|\mathbf{X} - \tilde{\mathbf{A}}\tilde{\mathbf{B}}\mathbf{X}\|_F^2$ 
12:    if  $\text{RSS}^{(t)} < (1 + \epsilon) \cdot \text{RSS}$  then
13:       $\mu \leftarrow 1.2 \cdot \mu$ 
14:       $\text{RSS} \leftarrow \text{RSS}^{(t)}$ 
15:      break
16:    else
17:       $\mu \leftarrow 0.5 \cdot \mu$ 
18:    end if
19:  end for
20: end for
21: Return:  $\tilde{\mathbf{B}}$ 

```

2.3.3 Frank-Wolfe (FW) Algorithm

The Frank-Wolfe (FW) algorithm is a first-order iterative method for solving constrained optimization problems. In contrast to projected-gradient methods, such as PCHA, FW replaces the projection step with a linear minimization over the feasible set. Specifically, it solves a linear approximation of the objective at each iterate and moves toward the minimizer, thereby ensuring that all updates remain within the feasible region. For the classical FW algorithm to be applicable, the feasible set must be compact and convex. The absence of projections makes FW particularly attractive for structured domains where linear minimization is computationally cheaper than projection. For more details, see (Bauckhage et al. 2015; Clarkson 2010; Frank and Wolfe 1956; Jaggi 2013).

In our alternating optimization scheme the optimization in $\mathbf{A}$ (resp. $\mathbf{B}$) is over the standard $(K - 1)$ -simplex (resp. $(N - 1)$ -simplex), both of which are convex (Appendix 6.5) and compact (Appendix 6.6). Thus the prerequisites for the vanilla FW algorithm are met.

Given a continuously differentiable objective function $f : \mathcal{D} \rightarrow \mathbb{R}$, and convex, compact feasible set $\mathcal{D}$, the FW algorithm is outlined in Algorithm 4.

Algorithm 4 Vanilla Frank-Wolfe (Frank and Wolfe 1956; Jaggi 2013)

- 1: **Input:** Convex, compact feasible set $\mathcal{D}$, continuously differentiable objective function $f : \mathcal{D} \rightarrow \mathbb{R}$, number of iterations T
- 2: **Initialize:** $\mathbf{x}^{(1)} \in \mathcal{D}$
- 3: **for** $t = 1$ to T **do**
- 4: Compute linear minimizer:

$$\mathbf{s}^{(t)} = \arg \min_{\mathbf{s} \in \mathcal{D}} \left(\nabla f \left(\mathbf{x}^{(t)} \right) \right)^T \mathbf{s} \quad (30)$$

- 5: Set step size:

$$\mu^{(t)} = \frac{2}{t+2} \quad (31)$$

- 6: Update $\mathbf{x}$:

$$\mathbf{x}^{(t+1)} = (1 - \mu^{(t)})\mathbf{x}^{(t)} + \mu^{(t)}\mathbf{s}^{(t)} \quad (32)$$

- 7: **end for**

- 8: **Return:** $\mathbf{x}^{(T+1)}$
-

Because $0 \leq \mu^{(t)} \leq 1$ and both $\mathbf{x}^{(t)}$ and $\mathbf{s}^{(t)}$ lie in $\mathcal{D}$, every iterate $\mathbf{x}^{(t+1)}$ remains in the feasible set by convexity.

Now, in the case of archetypal analysis, when updating the rows of $\mathbf{A}$, the feasible set is the standard simplex Δ_{K-1} :

$$\mathbf{a}_n = \arg \min_{\mathbf{a}_n \in \Delta_{K-1}} \|\mathbf{x}_n - \mathbf{Z}^T \mathbf{a}_n\|_2^2. \quad (33)$$

As shown in Equation 125, the gradient with respect to $\mathbf{a}_n$ is given by

$$g_n^{(\mathbf{A})} = 2 \left(\mathbf{a}_n^T \mathbf{Z} \mathbf{Z}^T - \mathbf{x}_n^T \mathbf{Z} \right). \quad (34)$$

Accordingly, the FW linear subproblem becomes

$$\mathbf{s}^{(t)} = \arg \min_{\mathbf{s} \in \Delta_{K-1}} \left(g_n^{(\mathbf{A})} \right)^T \mathbf{s}. \quad (35)$$

Since the objective is linear in $\mathbf{s}$ and Δ_{K-1} is convex and compact, the minimizer is attained at a vertex of the simplex. That is, the solution lies in $\{\mathbf{e}_1, \dots, \mathbf{e}_K\}$, where $\mathbf{e}_k$ denotes the k th standard basis vector:

$$\mathbf{s}^{(t)} = \arg \min_{\mathbf{s} \in \{\mathbf{e}_1, \dots, \mathbf{e}_K\}} \left(g_n^{(\mathbf{A})} \right)^T \mathbf{s}. \quad (36)$$

This simply selects the direction of steepest descent, i.e. the component where the gradient is most negative:

$$\mathbf{s}^{(t)} = \mathbf{e}_{k'} \quad \text{where} \quad k' = \arg \min_{k \in \{1, \dots, K\}} g_{nk}^{(\mathbf{A})}. \quad (37)$$

Putting everything together, we obtain the update rule for $\mathbf{A}$, as outlined in Algorithm 5.

Algorithm 5 Update $\mathbf{A}$ via Frank-Wolfe

- 1: **Input:** $\mathbf{X}, \mathbf{A}, \mathbf{Z}$
- 2: **Output:** Updated $\mathbf{A}$
- 3: Initialize: $\mathbf{A}^{(1)} = \mathbf{A}$
- 4: **for** $t = 1$ to T **do**
- 5: Compute gradient for all rows of $\mathbf{A}$:

$$\mathbf{G}^{(\mathbf{A})} \leftarrow 2(\mathbf{A}\mathbf{Z}\mathbf{Z}^T - \mathbf{X}\mathbf{Z}^T) \quad (38)$$

- 6: **for** $n = 1$ to N **do**
- 7: Compute linear minimizer:

$$\mathbf{s}_n^{(t)} = \mathbf{e}_{k'} \quad \text{where} \quad k' = \arg \min_{k \in \{1, \dots, K\}} g_{nk}^{(\mathbf{A})} \quad (39)$$

- 8: Set step size:

$$\mu^{(t)} = \frac{2}{t+2} \quad (40)$$

- 9: Update $\mathbf{a}_n$:

$$\mathbf{a}_n^{(t+1)} = (1 - \mu^{(t)})\mathbf{a}_n^{(t)} + \mu^{(t)}\mathbf{s}_n^{(t)} \quad (41)$$

- 10: **end for**
 - 11: **end for**
 - 12: **Return:** $\mathbf{A}^{(T+1)}$
-

Following the derivation of the FW update of $\mathbf{A}$, the FW update for $\mathbf{B}$ is outlined in Algorithm 6.

Algorithm 6 Update $\mathbf{B}$ via Frank-Wolfe

- 1: **Input:** $\mathbf{X}, \mathbf{A}, \mathbf{B}$
- 2: **Output:** Updated $\mathbf{B}$
- 3: Initialize: $\mathbf{B}^{(1)} = \mathbf{B}$
- 4: **for** $t = 1$ to T **do**
- 5: Compute gradient for all rows of $\mathbf{B}$:

$$\mathbf{G}^{(\mathbf{B})} \leftarrow 2(\mathbf{A}^T \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T - \mathbf{A}^T \mathbf{X} \mathbf{X}^T) \quad (42)$$

- 6: **for** $k = 1$ to K **do**
- 7: Compute linear minimizer:

$$\mathbf{s}_k^{(t)} = \mathbf{e}_{n'} \quad \text{where} \quad n' = \arg \min_{n \in \{1, \dots, N\}} g_{kn}^{(\mathbf{B})} \quad (43)$$

- 8: Set step size:

$$\mu^{(t)} = \frac{2}{t+2} \quad (44)$$

- 9: Update $\mathbf{b}_k$:

$$\mathbf{b}_k^{(t+1)} = (1 - \mu^{(t)})\mathbf{b}_k^{(t)} + \mu^{(t)}\mathbf{s}_k^{(t)} \quad (45)$$

- 10: **end for**
 - 11: **end for**
 - 12: **Return:** $\mathbf{B}^{(T+1)}$
-

2.4 Initialization

2.4.1 Uniform

Sampling data points with uniform probability from the data set to initialize the archetypes was the first initialization scheme used for archetypal analysis (Cutler and Breiman 1994). However, already this original work the authors stated that initializing archetypes too close to each other can impede convergence speed.

2.4.2 Furthest Sum

Inspired by the *FurthestFirst* initialization for K -means clustering (Hochbaum and Shmoys 1985), the *FurthestSum* initialization for archetypal analysis was introduced by Mørup and Hansen (2012). The algorithm is outlined in Algorithm 7. The idea is to greedily select points such that each point that is added, maximizes the sum of distances to all the other points that have already been chosen. To dampen the effect of an "unlucky" random seed, we let the selection run for $K + 10$ iterations and discard the first 10 indices, this "burn-in" leaves exactly K archetypes.

Algorithm 7 Furthest Sum Initialization

```

1: Input: Data matrix  $\mathbf{X} \in \mathbb{R}^{N \times D}$ , number of archetypes  $K \leq N$ 
2: Output: Set of selected indices  $\mathcal{S}$ 
3: Define full index set  $\mathcal{I} = \{1, \dots, N\}$ 
4: Initialize set of selected indices  $\mathcal{S} \leftarrow (i_1)$ , where  $i_1$  is randomly chosen
5: for  $t = 2$  to  $K + 10$  do
6:   if  $|\mathcal{S}| > K$  then
7:      $\mathcal{S} \leftarrow \mathcal{S} \setminus i_{t-K}$  ▷ Remove the oldest index from  $\mathcal{S}$ 
8:   end if
9:   for  $n \in \mathcal{I} \setminus \mathcal{S}$  do ▷ Compute distances to current archetypes
10:     $d_n^{(\text{sum})} = \sum_{s \in \mathcal{S}} \|x_n - x_s\|_2$ 
11:   end for
12:    $i_t = \arg \max_{n \in \mathcal{I} \setminus \mathcal{S}} d_n^{(\text{sum})}$  ▷ Select new data point
13:    $\mathcal{S} \leftarrow \mathcal{S} \cup \{i_t\}$  ▷ Add  $i_t$  to  $\mathcal{S}$ 
14: end for
15: Return:  $\mathcal{S}$ 

```

2.4.3 Archetypal Analysis++

The *FurthestSum* algorithm does not guarantee that the selected archetypes are non-redundant. An archetype is redundant if it lies in the convex hull of the already selected archetypes (Suleman 2017b). Furthermore, in some cases it has been reported that the *FurthestSum* initialization yields inferior results compared to the uniform initialization (Krohne et al. 2019; Olsen et al. 2022). To address these limitations, Mair and Sjölund (2024) introduced the *AA++* initialization algorithm outlined in Algorithm 8. At each iteration, the sampling probability is proportional to distance to the convex hull of the already selected archetypes, which ensures that new archetypes are not redundant.

Algorithm 8 Archetypal++ Initialization

```

1: Input: Data matrix  $\mathbf{X} \in \mathbb{R}^{N \times D}$ , number of archetypes  $K \leq N$ 
2: Output: Set of selected indices  $\mathcal{S}$ 
3: Initialize set of selected indices  $\mathcal{S} \leftarrow \{i_1\}$ , where  $i_1$  is randomly chosen
4: for  $t = 2$  to  $K$  do
5:   Let  $\mathbf{Z}$  be the submatrix of  $\mathbf{X}$  with rows in  $\mathcal{S}$ 
6:   for  $n = 1$  to  $N$  do
7:      $d_n = \min_{a_n} \|x_n - \mathbf{Z}^T a_n\|_2$  ▷ Compute distance to convex hull of  $\mathbf{Z}$ 
8:   end for
9:   for  $n = 1$  to  $N$  do
10:     $p_n = \frac{d_n}{\sum_{n'} d_{n'}}$  ▷ Compute sample probability
11:   end for
12:    $i_t \sim \text{Categorical}(\mathbf{p})$  ▷ Sample new data point
13:    $\mathcal{S} \leftarrow \mathcal{S} \cup \{i_t\}$  ▷ Add  $i_t$  to  $\mathcal{S}$ 
14: end for
15: Return:  $\mathcal{S}$ 

```

2.5 Coresets

The idea of a *coreset* is to replace the full dataset $\mathcal{X} = \{\mathbf{x}_n\}_{n=1}^N$ with a much smaller, *weighted* subset

$$\tilde{\mathcal{X}} := \{(w_{\tilde{n}} \in \mathbb{R}_+, \mathbf{x}_{\tilde{n}} \in \mathcal{X})\}_{\tilde{n}=1}^{\tilde{N}} \quad \text{where} \quad \tilde{N} \ll N \quad (46)$$

such that models fitted on the coreset also provide a good fit on the original dataset (Feldman 2020).

However there are many different *coreset* definitions, as reviewed by Feldman (2020). In the following, we focus on three specific classes of coresets. Consider an optimization problem of the form

$$\min_{\mathcal{Z} \in \Theta} \sum_{n=1}^N d(\mathbf{x}_n, \mathcal{Z})^2 \quad (47)$$

where $\Theta := \{\{z_1, \dots, z_{K'}\} \subset \mathbb{R}^D \mid K' \leq K\}$ denotes the set of candidate solution sets $\mathcal{Z} \subset \mathbb{R}^D$ of cardinality at most K , and d denotes some distance or loss function, for example in archetypal analysis (see Section 2.2.5) and K -means clustering:

$$d(\mathbf{x}_n, \mathcal{Z}) = \begin{cases} \min_{\mathbf{q} \in \text{conv}(\mathcal{Z})} \|\mathbf{x}_n - \mathbf{q}\|_2^2, & (\text{Archetypal Analysis}) \\ \min_{\mathbf{z} \in \mathcal{Z}} \|\mathbf{x}_n - \mathbf{z}\|_2^2, & (K\text{-means}) \end{cases} \quad (48)$$

Now, for $\varepsilon \in [0, 1]$ and $K \in \mathbb{N}$, a weighted subset $\tilde{\mathcal{X}} = \{(w_{\tilde{n}}, x_{\tilde{n}})\}_{\tilde{n}=1}^{\tilde{N}}$ with $x_{\tilde{n}} \in \mathcal{X}$ and $w_{\tilde{n}} \in \mathbb{R}_+$ is called an ε -coreset, ε -lightweight-coreset, or ε -absolute-coreset, respectively, if for all $\mathcal{Z} \in \Theta$,

$$\left| \sum_{n=1}^N d(\mathbf{x}_n, \mathcal{Z}) - \sum_{\tilde{n}=1}^{\tilde{N}} w_{\tilde{n}} d(\mathbf{x}_{\tilde{n}}, \mathcal{Z}) \right| \leq \begin{cases} \varepsilon \sum_{n=1}^N d(\mathbf{x}_n, \mathcal{Z}), & \varepsilon\text{-coreset}, \\ \frac{\varepsilon}{2} \sum_{n=1}^N d(\mathbf{x}_n, \mathcal{Z}) + \frac{\varepsilon}{2} \sum_{n=1}^N d(\mathbf{x}_n, \{\bar{\mathbf{x}}\}), & \varepsilon\text{-lightweight-coreset}, \\ \varepsilon, & \varepsilon\text{-absolute-coreset}. \end{cases} \quad (49)$$

The standard (relative) ε -coreset is included here only as a reference, as it is the variant most widely used in the coreset literature and provides a point of comparison with the lightweight and absolute forms. It

is usually written in the multiplicative form

$$\forall \mathcal{Z} \in \Theta: \quad (1 - \varepsilon) \sum_{n=1}^N d(\mathbf{x}_n, \mathcal{Z}) \leq \sum_{\tilde{n}=1}^{\tilde{N}} w_{\tilde{n}} d(\mathbf{x}_{\tilde{n}}, \mathcal{Z}) \leq (1 + \varepsilon) \sum_{n=1}^N d(\mathbf{x}_n, \mathcal{Z}) \quad (50)$$

which highlights the $(1 \pm \varepsilon)$ multiplicative approximation guarantees.

Mair and Brefeld (2019) showed that distance used in archetypal analysis is upper-bounded by the distance used in K -means clustering (see Lemma 1 and Proposition 1). Hence, the archetypal analysis loss is upper bounded by the K -means loss, i.e. for all $\mathcal{Z} \in \Theta$ we have

$$\sum_{n=1}^N \min_{\mathbf{z} \in \text{conv}(\mathcal{Z})} \|\mathbf{x}_n - \mathbf{z}\|_2^2 \leq \sum_{n=1}^N \min_{\mathbf{z} \in \mathcal{Z}} \|\mathbf{x}_n - \mathbf{z}\|_2^2 \quad (51)$$

Consequently, any ε -coreset for K -means also satisfies Equation 49 for AA, with the same ε . Mair and Brefeld (2019) then constructed the first ε -*absolute-coreset* for archetypal analysis by adapting the *lightweight-coreset* construction of Bachem, Lucic, and Krause (2018). Note that the coreset of Mair and Brefeld (2019) is not provably valid for every $\mathcal{Z} \in \Theta$; its guarantee holds only when the data mean $\bar{\mathbf{x}}$ lies in $\text{conv}(\mathcal{Z})$ (see Corollary 1). In practice this requirement is mild, because archetypes are extreme points and their convex hull typically encloses $\bar{\mathbf{x}}$. The resulting algorithm is outlined in Algorithm 9.

Algorithm 9 Coreset Construction for Archetypal Analysis

```

1: Input: Data matrix  $\mathbf{X} \in \mathbb{R}^{N \times D}$ , coreset size  $\tilde{N} < N$ 
2: Output: Coreset  $\tilde{\mathbf{X}}$ 
3:  $\bar{\mathbf{x}} \leftarrow \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$  ▷ Compute mean of data
4: for  $n = 1$  to  $N$  do
5:    $q(\mathbf{x}_n) = \frac{d(\mathbf{x}_n, \bar{\mathbf{x}})^2}{\sum_{n'=1}^N d(\mathbf{x}_{n'}, \bar{\mathbf{x}})^2}$  ▷ Compute sampling probability
6: end for
7:  $\tilde{\mathcal{X}} \leftarrow \emptyset$ 
8: for  $\tilde{n} = 1$  to  $\tilde{N}$  do
9:    $\tilde{\mathbf{x}}_{\tilde{n}} \sim q(\mathbf{x}_1, \dots, \mathbf{x}_N)$  ▷ Sample points with replacement
10:   $w_{\tilde{n}} = \frac{1}{\tilde{N} q(\tilde{\mathbf{x}}_{\tilde{n}})}$  ▷ Assign weight
11:   $\tilde{\mathcal{X}} \leftarrow \tilde{\mathcal{X}} \cup \{(w_{\tilde{n}}, \tilde{\mathbf{x}}_{\tilde{n}})\}$ 
12: end for
13: Return:  $\tilde{\mathcal{X}} = \{(w_{\tilde{n}}, \tilde{\mathbf{x}}_{\tilde{n}})\}_{\tilde{n}=1}^{\tilde{N}}$ 

```

Accordingly the archetypal analysis objective from Equation 4 is adapted to incorporate weights

$$\begin{aligned} \mathbf{A}, \mathbf{B} &= \arg \min_{\substack{\mathbf{A} \in F(\tilde{N}, K) \\ \mathbf{B} \in F(K, \tilde{N})}} \sum_{\tilde{n}=1}^{\tilde{N}} w_{\tilde{n}} \|\tilde{\mathbf{x}}_{\tilde{n}} - \tilde{\mathbf{X}}^T \mathbf{B}^T \mathbf{a}_{\tilde{n}}\|_2^2 \\ &= \arg \min_{\substack{\mathbf{A} \in F(\tilde{N}, K) \\ \mathbf{B} \in F(K, \tilde{N})}} \|\mathbf{W} \tilde{\mathbf{X}} - \mathbf{WAB} \tilde{\mathbf{X}}\|_F^2 \end{aligned} \quad (52)$$

where $\mathbf{W} := \text{diag}(w_1, \dots, w_{\tilde{N}})$ and $\tilde{\mathbf{X}} \in \mathbb{R}^{\tilde{N} \times D}$ denotes the subsampled data. Because the weights do not influence the update rule for $\mathbf{A}$ (since each $\mathbf{a}_n$ is updated independently), it suffices to replace $\mathbf{X}$ by $\tilde{\mathbf{X}}$. In contrast, the update rule for $\mathbf{B}$ must be adjusted to account for the weights; it depends on both $\tilde{\mathbf{X}}$ and the weighted matrix $\mathbf{W} \tilde{\mathbf{X}}$.

$$\mathbf{B} \leftarrow \arg \min_{\mathbf{B} \in F(K, \tilde{N})} \|\mathbf{W} \tilde{\mathbf{X}} - \mathbf{WAB} \tilde{\mathbf{X}}\|_F^2 \quad (53)$$

Originally the authors adapted the regularized nonnegative least squares (see Equation 19) by recasting the update for $\mathbf{B}$ as

$$\begin{aligned} \mathbf{Z} &\leftarrow \arg \min_{\mathbf{Z} \in \mathbb{R}^{K \times D}} \|\mathbf{W}\tilde{\mathbf{X}} - \mathbf{WAZ}\|_F^2 \\ \mathbf{b}_k &\leftarrow \arg \min_{\mathbf{b}_k \in \mathbb{R}^{\tilde{N}}} \left\| \begin{bmatrix} \mathbf{z}_k \\ \lambda \end{bmatrix} - \begin{bmatrix} \tilde{\mathbf{X}}^T \\ \lambda \mathbf{1}_{\tilde{N}}^T \end{bmatrix} \mathbf{b}_k \right\|_2^2 \end{aligned} \quad (54)$$

To fully capitalize on the computational speed-ups afforded by the coreset, we replace the NNLS update with the more efficient PCHA and FW algorithm. Accordingly, we re-derive their update rules from the weighted objective in Equation 53, using its gradient with respect to $\mathbf{B}$

$$\begin{aligned} \tilde{G}^{(B)} &= \nabla_B \widetilde{\text{RSS}} \\ &= \nabla_B \|\mathbf{W}\tilde{\mathbf{X}} - \mathbf{WAB}\tilde{\mathbf{X}}\|_F^2 \\ &= 2 \left((\mathbf{WA})^T (\mathbf{WA}) \mathbf{B} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^T - (\mathbf{WA})^T (\mathbf{W}\tilde{\mathbf{X}}) \tilde{\mathbf{X}}^T \right) \end{aligned} \quad (55)$$

To summarize, Algorithm 10 shows how to adapt a generic archetypal analysis solver to a coreset.

Algorithm 10 Prototypical Algorithm Optimization using Coreset

- 1: **Input:** Data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$, subsampled data matrix $\tilde{\mathbf{X}} \in \mathbb{R}^{\tilde{N} \times D}$, coreset weights $\mathbf{W} \in \mathbb{R}_+^{\tilde{N} \times \tilde{N}}$
number of archetypes K , max iterations T
- 2: **Initialize:** Archetypes $\mathbf{Z} \in \mathbb{R}^{K \times D}$
- 3: **for** $t = 1$ to T **do**
- 4: Compute optimal weights $\mathbf{A} \in \mathbb{R}^{N \times K}$:

$$\mathbf{A} \leftarrow \arg \min_{\mathbf{A} \in F(\tilde{N}, K)} \|\tilde{\mathbf{X}} - \mathbf{AZ}\|_F^2 \quad (56)$$

- 5: Compute optimal weights $\mathbf{B} \in \mathbb{R}^{K \times N}$:

$$\mathbf{B} \leftarrow \arg \min_{\mathbf{B} \in F(K, \tilde{N})} \|\mathbf{W}\tilde{\mathbf{X}} - \mathbf{WAB}\tilde{\mathbf{X}}\|_F^2 \quad (57)$$

- 6: Update archetypes:

$$\mathbf{Z} \leftarrow \mathbf{B}\tilde{\mathbf{X}} \quad (58)$$

- 7: **if** convergence criterion is met **then**
- 8: **break**
- 9: **end if**

- 10: **end for**

- 11: Recompute $\mathbf{A} \in F(N, K)$ on full dataset

$$\mathbf{A} \leftarrow \arg \min_{\mathbf{A} \in F(N, K)} \|\mathbf{X} - \mathbf{AZ}\|_F^2 \quad (59)$$

- 12: **Return:** $\mathbf{A}, \mathbf{B}, \mathbf{Z}$
-

In particular, only the following adaptations are required

1. Pre-compute the weighted data matrix, $\mathbf{W}\tilde{\mathbf{X}}$.
2. Evaluate the gradient with respect to $\mathbf{B}$ using Equation (55).
3. After the algorithm converges, optimize $\mathbf{A}$ on the full dataset $\mathbf{X}$ while holding $\mathbf{Z} = \mathbf{B}\tilde{\mathbf{X}}$ fixed.

2.6 Relaxation of Archetype Constraints

We can relax the constraint that archetypes must be convex combinations of data points (i.e. that they must lie within the convex hull of the data).

$$\begin{aligned} \hat{\mathbf{A}}, \hat{\mathbf{B}} = \arg \min_{\substack{\mathbf{A} \in \mathbb{R}^{N \times K} \\ \mathbf{B} \in \mathbb{R}^{K \times N}} \|\mathbf{X} - \mathbf{ABX}\|_F^2 \quad \text{subject to} \\ \mathbf{A} \geq 0, \quad \mathbf{A}\mathbf{1}_K = \mathbf{1}_N \\ \mathbf{B} \geq 0, \quad \forall k \in \{1, \dots, K\} \quad 1 - \delta \leq \|\mathbf{b}_k\|_1 \leq 1 + \delta \end{aligned} \quad (60)$$

Note that changing the constraint on $\mathbf{B}$ into two inequality constraints, the objective is still convex in $\mathbf{B}$ given a fixed $\mathbf{A}$ (Mørup and Hansen 2012). To optimize Equation 60, Mørup and Hansen (2012) first rewrite the objective by introducing a scaling vector $\boldsymbol{\alpha} \in \mathbb{R}_+^K$

$$\begin{aligned} \hat{\mathbf{A}}, \hat{\mathbf{B}} = \arg \min \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{BX}\|_F^2 \quad \text{subject to} \\ \mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N) \\ \forall k \in \{1, \dots, K\} \quad 1 - \delta \leq \alpha_k \leq 1 + \delta \end{aligned} \quad (61)$$

such that each row k of $\mathbf{B}$ is scaled by α_k . This formulation readily extends both the PCHA and the FW algorithm to the relaxed objective. Specifically, we can apply either method to update $\mathbf{A}$ and $\mathbf{B}$, using the gradients derived in Section 6.9 and shown below. The gradient with respect to $\mathbf{A}$ is structurally identical to the standard case, with the archetype matrix given by $\mathbf{Z} = \text{diag}(\boldsymbol{\alpha})\mathbf{BX}$.

$$\begin{aligned} \mathbf{G}^{(A)} &= \nabla_{\mathbf{A}} \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{BX}\|_F^2 \\ &= 2 (\mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{BXX}^T \mathbf{B}^T \text{diag}(\boldsymbol{\alpha}) - \mathbf{XX}^T \mathbf{B}^T \text{diag}(\boldsymbol{\alpha})) \\ &= 2 (\mathbf{AZZ}^T - \mathbf{XZ}^T) \end{aligned} \quad (62)$$

For $\mathbf{B}$ the gradient is given by

$$\begin{aligned} \mathbf{G}^{(B)} &= \nabla_{\mathbf{B}} \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{BX}\|_F^2 \\ &= 2 [\text{diag}(\boldsymbol{\alpha}) \mathbf{A}^T \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{BXX}^T - \text{diag}(\boldsymbol{\alpha}) \mathbf{A}^T \mathbf{XX}^T] \end{aligned} \quad (63)$$

At each iteration, $\boldsymbol{\alpha}$ can be updated via projected gradient descent using the gradient

$$\begin{aligned} G^{(\alpha)} &= \nabla_{\alpha} \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{BX}\|_F^2 \\ &= 2 [\mathbf{A}^T \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{BXX}^T \mathbf{B}^T - \mathbf{A}^T \mathbf{XX}^T \mathbf{B}^T] \end{aligned} \quad (64)$$

and the component-wise projection onto $[1 - \delta, 1 + \delta]$.

The algorithm is outlined in Algorithm 11.

Algorithm 11 Prototypical Algorithm for Objective with Relaxed Archetypal Constraints

-
- 1: **Input:** Data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$, number of archetypes K , max iterations T
 - 2: **Initialize:** Archetypes $\mathbf{Z} \in \mathbb{R}^{K \times D}$, $\boldsymbol{\alpha} = \mathbf{1}_K$
 - 3: **for** $t = 1$ to T **do**
 - 4: Compute optimal weights $\mathbf{A} \in \mathbb{R}^{N \times K}$:

$$\mathbf{A} \leftarrow \arg \min_{\mathbf{A} \in F(N, K)} \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \mathbf{X}\|_F^2 \quad (65)$$

- 5: Compute optimal weights $\mathbf{B} \in \mathbb{R}^{K \times N}$:

$$\mathbf{B} \leftarrow \arg \min_{\mathbf{B} \in F(K, N)} \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \mathbf{X}\|_F^2 \quad (66)$$

- 6: Compute optimal $\boldsymbol{\alpha} \in [1 - \delta, 1 + \delta]^K$:

$$\boldsymbol{\alpha} \leftarrow \arg \min_{\boldsymbol{\alpha} \in [1 - \delta, 1 + \delta]^K} \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \mathbf{X}\|_F^2 \quad (67)$$

- 7: Update archetypes:

$$\mathbf{Z} \leftarrow \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \mathbf{X} \quad (68)$$

- 8: **if** convergence criterion is met **then**
 - 9: **break**
 - 10: **end if**
 - 11: **end for**
 - 12: **Return:** $\mathbf{A}, \mathbf{B}, \mathbf{Z}$
-

2.7 Combining Coresets and Relaxation of Archetype Constraints

If we seek to use coresets and relax the constraint that archetypes must lie within the convex hull of the data, then we need to optimize the objective

$$\begin{aligned} \hat{\mathbf{A}}, \hat{\mathbf{B}} &= \arg \min \|\mathbf{W} \tilde{\mathbf{X}} - \mathbf{W} \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \tilde{\mathbf{X}}\|_F^2 \quad \text{subject to} \\ \mathbf{A} &\in F(\tilde{N}, K) \\ \mathbf{B} &\in F(K, \tilde{N}) \\ \forall k \in \{1, \dots, K\} \quad &1 - \delta \leq \alpha_k \leq 1 + \delta \end{aligned} \quad (69)$$

As in Section 2.6, we can readily extend both the PCHA and the FW algorithm to the objective in Equation 69. Specifically, to update $\mathbf{A}$ we can simply use the gradient shown in Equation 62. To update $\mathbf{B}$ and $\boldsymbol{\alpha}$ we need to derive their gradient with respect to Equation 69. For $\mathbf{B}$ we have

$$\begin{aligned} G^{(B)} &= \nabla_{\mathbf{B}} \text{RSS} \\ &= 2 \left[\text{diag}(\boldsymbol{\alpha}) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^T - \text{diag}(\boldsymbol{\alpha}) \check{\mathbf{A}}^T \check{\mathbf{X}} \tilde{\mathbf{X}}^T \right] \end{aligned} \quad (70)$$

as derived in Equation 143. For $\boldsymbol{\alpha}$ we have

$$\begin{aligned} G^{(\alpha)} &= \nabla_{\boldsymbol{\alpha}} \text{RSS} \\ &= 2 \left[\check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \tilde{\mathbf{X}} \tilde{\mathbf{X}}^T \mathbf{B}^T - \check{\mathbf{A}}^T \check{\mathbf{X}} \tilde{\mathbf{X}}^T \mathbf{B}^T \right] \end{aligned} \quad (71)$$

as derived in Equation 144. The corresponding algorithm is outlined in Algorithm 11

Algorithm 12 Prototypical Algorithm for using Coresets and Relaxation of Archetypal Constraints

-
- 1: **Input:** Data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$, Subsampled data matrix $\tilde{\mathbf{X}} \in \mathbb{R}^{\tilde{N} \times D}$, coreset weights $\mathbf{W} \in \mathbb{R}_+^{\tilde{N} \times \tilde{N}}$ number of archetypes K , max iterations T
 - 2: **Initialize:** Archetypes $\mathbf{Z} \in \mathbb{R}^{K \times D}$, $\boldsymbol{\alpha} = \mathbf{1}_K$
 - 3: **for** $t = 1$ to T **do**
 - 4: Compute optimal weights $\mathbf{A} \in \mathbb{R}^{N \times K}$:

$$\mathbf{A} \leftarrow \arg \min_{\mathbf{A} \in F(\tilde{N}, K)} \|\tilde{\mathbf{X}} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \tilde{\mathbf{X}}\|_F^2 \quad (72)$$

- 5: Compute optimal weights $\mathbf{B} \in \mathbb{R}^{K \times N}$:

$$\mathbf{B} \leftarrow \arg \min_{\mathbf{B} \in F(K, \tilde{N})} \|\mathbf{W} \tilde{\mathbf{X}} - \mathbf{W} \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \tilde{\mathbf{X}}\|_F^2 \quad (73)$$

- 6: Compute optimal $\boldsymbol{\alpha} \in [1 - \delta, 1 + \delta]^K$:

$$\boldsymbol{\alpha} \leftarrow \arg \min_{\boldsymbol{\alpha} \in [1 - \delta, 1 + \delta]^K} \|\mathbf{W} \tilde{\mathbf{X}} - \mathbf{W} \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \tilde{\mathbf{X}}\|_F^2 \quad (74)$$

- 7: Update archetypes:

$$\mathbf{Z} \leftarrow \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \tilde{\mathbf{X}} \quad (75)$$

- 8: **if** convergence criterion is met **then**

- 9: **break**

- 10: **end if**

- 11: **end for**

- 12: Recompute $\mathbf{A} \in F(N, K)$ on full dataset

$$\mathbf{A} \leftarrow \arg \min_{\mathbf{A} \in F(N, K)} \|\mathbf{X} - \mathbf{A} \mathbf{Z}\|_F^2 \quad (76)$$

- 13: **Return:** $\mathbf{A}, \mathbf{B}, \mathbf{Z}$
-

2.8 Simulating Archetypes

2.8.1 Archetype Generation

Let K be the number of archetypes and D the dimensionality of the feature space. We first generate a set of M candidate points $C = \{c_1, \dots, c_M\} \subset \mathbb{R}^D$, sampled independently from a standard multivariate normal

$$\mathbf{c}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D) \quad (77)$$

We compute the convex hull $\text{conv}(C)$ of the candidate set and extract its vertices:

$$\mathcal{V} = \text{vertices}(\text{conv}(C)) \subseteq C. \quad (78)$$

Thereby, we make sure that we will not have any redundant archetypes. Note however that the procedure is computationally tractable only for modest dimensions D ; in high-dimensional spaces the convex-hull construction incurs a worst-case complexity of $\mathcal{O}(M^{\lfloor D/2 \rfloor})$, making the overall algorithm prohibitively expensive.

If $|\mathcal{V}| \geq K$, we select a subset $Z = \{\mathbf{z}_1, \dots, \mathbf{z}_K\} \subset \mathcal{V}$ by iteratively selecting points that maximize the minimum Euclidean distance to any of the points already selected. I.e. let $\mathcal{S} = \{\mathbf{v}_1, \dots, \mathbf{v}_{t-1}\}$ be the set

of points that have already been selected, then the next point is sampled according to:

$$\mathbf{v}_t = \arg \max_{\mathbf{v} \in \mathcal{V}} \min_{\mathbf{v}' \in \mathcal{S}} \|\mathbf{v} - \mathbf{v}'\|_2^2 \quad (79)$$

If $|\mathcal{V}| < K$, additional candidates are generated, and the process is repeated up to a maximum number of attempts.

2.8.2 Coefficient Matrix Sampling

Given N desired samples, we generate a matrix $\mathbf{A} \in \mathbb{R}^{N \times K}$ of mixing coefficients, where each row $\mathbf{a}_n = [a_{n1}, \dots, a_{nK}]^T$ is sampled from a Dirichlet distribution

$$\mathbf{a}_n \sim \text{Dirichlet}(\mathbf{1}_K) \quad (80)$$

ensuring that each $\mathbf{a}_n \in \Delta_{K-1}$.

2.8.3 Data Generation

Let $\mathbf{Z} \in \mathbb{R}^{K \times D}$ be the matrix whose rows are the selected archetypes. The synthetic data matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$ is constructed via convex combinations of the archetypes:

$$\mathbf{X} = \mathbf{AZ}. \quad (81)$$

Optionally, zero-mean isotropic Gaussian noise can be added to simulate measurement noise:

$$x_{nd} \leftarrow x_{nd} + \varepsilon_{nd}, \quad \varepsilon_{nd} \sim \mathcal{N}(0, \sigma^2), \quad (82)$$

where σ is a user-defined noise standard deviation.

2.9 Implementation Details

2.9.1 Centering & Scaling

By default we center the data matrix $\mathbf{X}$, and globally scale $\mathbf{X}$ by

$$\lambda = \frac{\|\mathbf{X}\|_F}{ND}. \quad (83)$$

This stabilizes the optimization, while translation and scaling invariance of the objective (see Section 2.2.1 and Section 2.2.2) ensure that the optimal $\mathbf{A}, \mathbf{B}$ are unaffected by this preprocessing.

2.9.2 Convergence

Regardless of the objective and optimization algorithm, we use the moving average of the relative change in RSS, defined at iteration $t > 0$, as:

$$c_t = \frac{1}{\min(t, 20)} \sum_{\tau=1}^{\min(t, 20)} \frac{\text{RSS}_{t-\tau} - \text{RSS}_{t-(\tau+1)}}{\text{RSS}_{t-(\tau+1)}} \quad (84)$$

We terminate the optimization if $c_t \geq 0$ or $|c_t| < 10^{-4}$.

2.9.3 Multiple Restarts

By default, the `compute_archetypes` function runs the optimization five times with different random seeds, retaining only the result with the lowest RSS.

3 Number of Archetypes

There is no universally optimal method for selecting the number of archetypes (Alcacer et al. 2025). Here, we provide the following heuristics to guide the decision.

3.1 Variance Explained

For a given number of archetypes, the variance explained is computed as

$$\begin{aligned} R^2 &= \frac{\text{TSS} - \text{RSS}}{\text{TSS}} \\ &= 1 - \frac{\text{RSS}}{\text{TSS}} \\ &= 1 - \frac{\|\mathbf{X} - \mathbf{ABX}\|_F^2}{\|\mathbf{X}\|_F^2} \end{aligned} \tag{85}$$

where $\mathbf{A}, \mathbf{B}$ denote the optimized coefficient matrices.

An "elbow" in this curve indicates that adding another archetype does not capture substantially more variance (Cutler and Breiman 1994; Eugster and Leisch 2009; Mørup and Hansen 2012).

3.2 Information-Theoretic Criterion

Suleman (2017c) adapted the information-theoretic criterion from Suleman (2017a) for archetypal analysis, defining it as

$$v_{AA} = \log \left(\frac{1}{ND} \|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 \right) + 2 \frac{K_a + K_b + 1}{N \text{tr}(\Sigma_{\hat{\mathbf{X}}} \Sigma_{\mathbf{X}}^{-1})} \tag{86}$$

where $\hat{\mathbf{X}} = \mathbf{ABX}$ is the reconstructed data matrix; $K_a = N(K - 1)$ (denoted K_μ in (Suleman 2017c)) is the number of parameters in $\mathbf{A}$; and $K_b = K(N - 1)$ (denoted K_β in (Suleman 2017c)) is the number of parameters in $\mathbf{B}$. The matrices $\Sigma_{\hat{\mathbf{X}}}$ and $\Sigma_{\mathbf{X}}$ denote the empirical covariances of $\hat{\mathbf{X}}$ and $\mathbf{X}$, respectively.

This criterion measures the goodness of fit by balancing model complexity and reconstruction accuracy: a lower value indicates a more favorable trade-off between the number of archetypes and the variance explained.

3.3 Bootstrapping

Following the approach in (Hart et al. 2015; Korem et al. 2015), we assess the stability of the inferred archetypes using a bootstrap-based method.

For a given number of archetypes K , we first compute the optimal archetypes $\{\mathbf{z}_1, \dots, \mathbf{z}_K\}$ on the full dataset.

Then, for each bootstrap sample $t = 1, \dots, T$, we compute the corresponding optimal archetypes $\{\mathbf{z}_1^{(t)}, \dots, \mathbf{z}_K^{(t)}\}$. These are matched to the original archetypes by solving the linear assignment problem:

$$\sigma^{(t)} = \arg \min_{\sigma \in S_K} \sum_{k=1}^K c(\mathbf{z}_k, \mathbf{z}_{\sigma(k)}^{(t)}), \quad (87)$$

where S_K is the set of all permutations of $\{1, \dots, K\}$, and the cost function c is defined as the Euclidean distance, i.e.,

$$c(\mathbf{z}_k, \mathbf{z}_{k'}) = \|\mathbf{z}_k - \mathbf{z}_{k'}\|_2. \quad (88)$$

We then compute the bootstrap variance per archetype k as

$$v_k = \frac{1}{TD} \sum_{t=1}^T \left\| \mathbf{z}_k^{(t)} - \bar{\mathbf{z}}_k \right\|_2^2, \quad (89)$$

where the mean archetype is defined as $\bar{\mathbf{z}}_k = \frac{1}{T} \sum_{t=1}^T \mathbf{z}_k^{(t)}$.

The variance v_k serves as a measure of the stability of archetype k : a low variance indicates a robust and well-defined archetype, whereas a high variance suggests sensitivity to resampling.

4 Archetype Characterization

To characterize each archetype k , we define a smooth weighting over all cells n using a squared exponential kernel:

$$w_{kn} = \frac{\exp \left[-\frac{\|\mathbf{z}_k - \mathbf{x}_n\|_2^2}{2\ell^2} \right]}{\sum_{n'=1}^N \exp \left[-\frac{\|\mathbf{z}_k - \mathbf{x}_{n'}\|_2^2}{2\ell^2} \right]}, \quad \mathbf{W} \in [0, 1]^{K \times N} \quad (90)$$

Thereby, cells that are close to an archetype get a large weight for that archetype.

By default, the length scale ℓ is determined automatically as

$$\ell = \frac{1}{2} \text{median}(\{\|\mathbf{z}_k - \bar{\mathbf{x}}\|_2 \mid k \in \{1, \dots, K\}\}) \quad (91)$$

where $\bar{\mathbf{x}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$ denotes the data mean.

Note that the weight computation also implies that for each archetype k we have

$$\mathbf{w}_k \geq 0 \text{ and } \sum_{n=1}^N w_{kn} = 1. \quad (92)$$

Multiplying the weight matrix $\mathbf{W} \in \mathbb{R}^{K \times N}$ with the original feature matrix $\mathbf{Y}^{(\mathbf{X})} \in \mathbb{R}^{N \times G}$ (where G denotes the number of original features) yields a feature profile for each archetype.

$$\mathbf{Y}^{(\mathbf{Z})} = \mathbf{W} \mathbf{Y}^{(\mathbf{X})} \in \mathbb{R}^{K \times G} \quad (93)$$

For downstream analysis, we multiply the weight matrix with the z-scored, log1p-normalized expression matrix $\mathbf{Y}_{\text{z-scores}}^{(\mathbf{X})}$ (where z-scores are computed across cells). This highlights genes and biological processes that distinguish the archetypes, since uniformly expressed genes exhibit low z-scores, while cell state-specific genes display high z-scores.

Similarly, to associate continuous covariates (e.g. age, pseudotime) with archetypes, we multiply the weight matrix with the covariate vector.

For categorical covariates, we first one-hot encode the covariate yielding some matrix $\mathbf{H} \in \{0, 1\}^{N \times C}$ where C is the number of categories. To account for different number of samples per category, we normalize each column (i.e. category) c to sum to one

$$\tilde{h}_{nc} = \frac{h_{nc}}{\sum_{n'=1}^N h_{n'c}}, \quad \tilde{\mathbf{H}} \in [0, 1]^{N \times C} \quad (94)$$

This normalization ensures that archetype scores are not biased by class size. Then multiplying the weight matrix $\mathbf{W}$ with $\tilde{\mathbf{H}}$ yields enrichment scores per archetype per category.

$$\mathbf{S} = \mathbf{W}\tilde{\mathbf{H}} \in \mathbb{R}_+^{K \times C} \quad (95)$$

4.1 Enrichment Analysis

To perform enrichment analysis we use decoupler-py (Badia-i-Mompel et al. 2022). By default, we use the univariate linear model (ULM) implemented in the package. Briefly, given some gene set with weights $\mathbf{w} = [w_1, \dots, w_G]^T$, the ULM method models the expression profile of archetype k , denoted $\mathbf{y}_k^{(Z)}$, as

$$\mathbf{y}_k^{(Z)} = \beta_0 + \beta_1 \mathbf{w} + \epsilon_k. \quad (96)$$

Intuitively, if genes with higher weights tend to be more highly expressed in archetype k , the coefficient β_1 will be large. Conversely, if the weights are unrelated to expression, β_1 will be close to zero.

For unweighted gene sets, such as the Hallmark gene sets (Subramanian et al. 2005), weights are binary—set to one if a gene is in the set and zero otherwise.

The enrichment score is then calculated as the t-value of β_1

$$t_{\beta_1} = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} \quad (97)$$

Note that the modular implementation of ParTIPy facilitates using any other enrichment analysis software.

4.2 Spatial Mapping

Consider the setting where we have a *dissociated* single-cell dataset with log1p-normalized gene-expression matrix $\mathbf{Y} \in \mathbb{R}^{N \times G}$ and corresponding low-dimensional embedding $\mathbf{X} \in \mathbb{R}^{N \times D}$. In the same biological context, we also have a spatial single-cell dataset with log1p-normalized expression matrix $\mathbf{Y}' \in \mathbb{R}^{N' \times G'}$, where typically $G' < G$. Because the dissociated dataset captures more genes and therefore more information, we infer the archetypes on $\mathbf{X}$, yielding the characteristic expression profiles $\mathbf{Y}^{(Z)} \in \mathbb{R}^{K \times G}$ as described above. To relate the spatial cells to these archetypes, we restrict $\mathbf{Y}^{(Z)}$ to the intersection of

genes between the two datasets (denoting the size of the intersection with G''), obtaining $\mathbf{Y}_G^{(\mathbf{Z})} \in \mathbb{R}^{K \times G''}$. We then apply decoupler-py’s univariate linear model (ULM) (Badia-i-Mompel et al. 2022) to each profile $\mathbf{y}_{k,G}^{(\mathbf{Z})}$ to compute enrichment scores which quantify how strongly each cell n' in the spatial dataset is associated with each archetype k .

4.3 Archetype Crosstalk Networks

The observed distribution of cells between archetypes can be understood as functional *division of labor* between the task that each archetype is specialized in. Ligand-receptor (LR) signaling (e.g. lateral inhibition) is one potential mechanism that establishes and maintains this pattern of specialization (Adler et al. 2023). To map the possible lines of communication between archetypes, we combine their characteristic expression profiles with LR databases (Dimitrov et al. 2022; Dimitrov et al. 2024), following the workflow adapted from Adler et al. (2023).

Step 1: Identify archetype-enriched genes. For each archetype k and gene g we compute a specificity score

$$s_{kg} = y_{kg}^{(\mathbf{Z})} - \max_{k' \neq k} y_{k'g}^{(\mathbf{Z})}, \quad (98)$$

where $\mathbf{Y}^{(\mathbf{Z})} \in \mathbb{R}^{K \times G}$ is the matrix of characteristic expression profiles inferred using the z-scored log1p single-cell expression profiles. Genes with $s_{kg} > \tau$ (default $\tau = 0.1$) are deemed *enriched* in archetype k .

Step 2: Filter to archetype-specific LR pairs. Let $\mathcal{D} = \{(l, r)\}$ be a curated set of cognate LR pairs (Dimitrov et al. 2022; Dimitrov et al. 2024). For every ordered pair of archetype (k, k') , we retain only those pairs whose ligand is enriched in k and whose receptor is enriched in k' :

$$\mathcal{D}_{k \rightarrow k'} = \{(l, r) \in \mathcal{D} : l \text{ enriched in } k, r \text{ enriched in } k'\}. \quad (99)$$

Step 3: Build archetype crosstalk network For every ordered pair (k, k') with $|\mathcal{D}_{k \rightarrow k'}| > 0$ we insert a directed edge $k \rightarrow k'$ and assign it the weight $w_{k \rightarrow k'} = |\mathcal{D}_{k \rightarrow k'}|$.

The resulting weighted graph summarizes the potential signaling axes that may coordinate the division of labor among archetypes.

4.4 Quantile-Based Archetype Enrichment

This enrichment strategy follows the distance-ranking framework introduced in (Adler et al. 2019; Hart et al. 2015; Korem et al. 2015) and is implemented following the ParTI MATLAB reference implementation (`ContinuousEnrichment.m`, `DiscreteEnrichment.m`; commit `f51c5af896bbd12c8da0e1104d3ff7557bea0c30`), serving as a complementary approach to the enrichment method described above.

For a fixed archetype k , we first compute the Euclidean distance between the archetype position $\mathbf{z}_k$ and the low-dimensional representation $\mathbf{x}_n$ of each cell n ,

$$d_{kn} = \|\mathbf{z}_k - \mathbf{x}_n\|_2. \quad (100)$$

Cells are then ordered by increasing distance to the archetype, and the resulting distances are transformed into ranks. This rank transformation ensures robustness to the absolute scale of distances.

Next, the ranked cells are partitioned into disjoint quantile bins of approximately equal size. Each bin contains a fixed fraction of the cells (by default 10%), with the exact bin size depending on the total number of cells. Larger datasets permit finer stratification of distances, while maintaining sufficiently large bin sizes to control estimator variance and ensure statistical robustness.

Enrichment is evaluated with respect to a distance-based contrast between subsets of quantile bins, which specifies the foreground (cells closest to the archetype) and a background (by default all remaining cells, but alternative contrasts such as closest versus furthest bins are also supported).

Continuous features. For continuous features such as gene expression, enrichment under the specified distance-based contrast is evaluated using summary statistics of expression across quantile bins. For each gene, summary statistics (default: median) are computed across all bins, and features are required to exhibit maximal expression in the bin closest to the archetype. Statistical significance of enrichment is then assessed by contrasting the foreground and background cell sets using a nonparametric rank-based test (Mann–Whitney U; default) (Mann and Whitney 1947), with the option to alternatively apply Welch’s two-sample t-test (Welch 1947). Multiple testing across features and archetypes is controlled using the Benjamini-Hochberg procedure (Benjamini and Hochberg 1995).

Discrete features. For discrete or categorical features (e.g. disease labels, sex), enrichment under the same distance-based contrast is assessed by analyzing the distribution of categories across distance-based quantile bins. For a feature with multiple categories, we treat each category level as a separate one-versus-rest indicator and test its over-representation near each archetype. For each archetype and category, we compute bin-wise category frequencies and an enrichment curve defined as the ratio between the category frequency within each bin and its global frequency across all cells. Categories are required to exhibit maximal enrichment in the bin closest to the archetype. Statistical significance of over-representation in the foreground relative to the background is evaluated using a hypergeometric test, and multiple testing across categories and archetypes is controlled using the Benjamini–Hochberg procedure.

This quantile-based formulation provides a simple and interpretable framework for archetype-specific enrichment analysis that applies uniformly to both continuous and discrete features, directly linking feature specificity to distance from archetypal states while avoiding assumptions on the functional form of feature gradients.

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6 Appendix

6.1 Proof for Translation Invariance

Let $\mathbf{v} \in \mathbb{R}^D$, and let $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{1}_N \mathbf{v}^T$ be the translated matrix. Then for any feasible $\mathbf{A}, \mathbf{B}$

$$\begin{aligned}\tilde{\mathbf{X}} - \mathbf{A}\mathbf{B}\tilde{\mathbf{X}} &= (\mathbf{X} + \mathbf{1}_N \mathbf{v}^T) - \mathbf{A}\mathbf{B}(\mathbf{X} + \mathbf{1}_N \mathbf{v}^T) \\ &= \mathbf{X} + \mathbf{1}_N \mathbf{v}^T - \mathbf{A}\mathbf{B}\mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{1}_N \mathbf{v}^T\end{aligned}\quad (101)$$

Since $\mathbf{B}\mathbf{1}_N = \mathbf{1}_K$ and $\mathbf{A}\mathbf{1}_K = \mathbf{1}_N$, this simplifies to

$$\begin{aligned}\tilde{\mathbf{X}} - \mathbf{A}\mathbf{B}\tilde{\mathbf{X}} &= \mathbf{X} + \mathbf{1}_N \mathbf{v}^T - \mathbf{A}\mathbf{B}\mathbf{X} - \mathbf{1}_N \mathbf{v}^T \\ &= \mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{X}\end{aligned}\quad (102)$$

Therefore, the reconstruction error remains unchanged, and the minimizers $\mathbf{A}, \mathbf{B}$ are invariant under such translations. Thus, the minimizers $\mathbf{A}, \mathbf{B}$ are invariant to centering the data.

6.2 Proof for Scaling Invariance

Let $\lambda \neq 0$, and let $\tilde{\mathbf{X}} = \lambda \mathbf{X}$ be the scaled matrix. Then for any feasible $\mathbf{A}, \mathbf{B}$

$$\begin{aligned}\tilde{\mathbf{X}} - \mathbf{A}\mathbf{B}\tilde{\mathbf{X}} &= \lambda \mathbf{X} - \mathbf{A}\mathbf{B}\lambda \mathbf{X} \\ &= \lambda (\mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{X})\end{aligned}\quad (103)$$

Thus the objective for the scaled matrix is given by

$$\arg \min_{\substack{\mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N)}} \|\tilde{\mathbf{X}} - \mathbf{A}\mathbf{B}\tilde{\mathbf{X}}\|_F^2 = \arg \min_{\substack{\mathbf{A} \in F(N, K) \\ \mathbf{B} \in F(K, N)}} \lambda^2 \|\mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{X}\|_F^2 \quad (104)$$

Since $\lambda \neq 0$, we have $\lambda^2 > 0$, and thus the objective is scaled by a positive constant. Multiplying the objective function by a positive scalar does not affect the location of its minimum, because the ordering of objective values is preserved. In particular, the first-order (stationarity) and second-order (convexity/curvature) necessary conditions for optimality remain unchanged under such scaling.

6.3 Proof for Uniqueness up to Permutation

Assuming that for each archetype k , there exists one data point n that is best reconstructed using only this archetype, i.e.

$$\forall k \in \{1, \dots, K\} \exists n \in \{1, \dots, N\} a_{nk} = 1 \quad (105)$$

and that for each archetype there exists one data point that is only used to define this archetype and not any other archetype, i.e.

$$\forall k \in \{1, \dots, K\} \exists n \in \{1, \dots, N\} b_{kn} > 0 \wedge b_{k'n} = 0 \forall k' \neq k \quad (106)$$

then the objective does not suffer from rotational ambiguity. I.e. any orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{K \times K}$ that satisfies

$$\begin{aligned}\|\mathbf{X} - \mathbf{A}\mathbf{B}\mathbf{X}\|_F^2 &= \|\mathbf{X} - \mathbf{A}\mathbf{Q}\mathbf{Q}^{-1}\mathbf{B}\mathbf{X}\|_F^2 \\ &= \|\mathbf{X} - \tilde{\mathbf{A}}\tilde{\mathbf{B}}\mathbf{X}\|_F^2\end{aligned}\quad (107)$$

must be a permutation matrix.

Condition 1 implies that $\mathbf{A}$ has at least K rows that have only one non-zero value, and this non-zero value equals one.

Condition 2 implies that $\mathbf{B}$ has at least K columns that have only one non-zero value.

Example: If we had 3 archetypes, the following weight matrices $\mathbf{A}$, $\mathbf{B}$ would satisfy these conditions

$$\mathbf{A} = \begin{bmatrix} \vdots & \vdots & \vdots \\ 1 & 0 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} \cdots & 0 & \cdots & 0.4 & \cdots & 0 & \cdots \\ \cdots & 0 & \cdots & 0 & \cdots & 0.1 & \cdots \\ \cdots & 0.2 & \cdots & 0 & \cdots & 0 & \cdots \end{bmatrix} \quad (108)$$

Let $\mathbf{Q} \in \mathbb{R}^{K \times K}$ be some invertible matrix

$$\mathbf{ABX} = \mathbf{AQQ}^{-1}\mathbf{BX} = \tilde{\mathbf{A}}\tilde{\mathbf{B}}\mathbf{X} \quad (109)$$

Requiring that $\tilde{\mathbf{A}} \in F(N, K)$ and $\tilde{\mathbf{B}} \in F(K, N)$ (i.e. that both $\tilde{\mathbf{A}}$, $\tilde{\mathbf{B}}$ are still row-stochastic), we can derive the following properties that $\mathbf{Q}$ must fulfill.

First, since we require $\tilde{\mathbf{A}} \geq 0$, and $\mathbf{A} \geq 0$, and $\tilde{\mathbf{A}} = \mathbf{AQQ}^{-1}$, and Equation (9) must hold, we know that $\mathbf{Q} \geq 0$.

Proof: Any column $k \in \{1, \dots, K\}$ of $\mathbf{QA}$ is given by a linear combination of the columns of $\mathbf{A}$

$$(\mathbf{QA})_{:,k} = \mathbf{A}q_{:,k} = \sum_{k'=1}^K a_{:,k}q_{k',k} \quad (110)$$

Now if any coefficient $q_{1,k}, \dots, q_{K,k}$ is negative, then the resulting vector must have at least one negative element, since according to condition one, for every column vector there exists some index $n' \in \{1, \dots, N\}$ where the value corresponds to one, and for all other column vectors, the value at that index is zero. (see also the example matrices above)

However, since we require that $\sum_{k'=1}^K a_{:,k}q_{k',k} \geq 0$, no element of $q_{:,k}$ can be negative, and this applies to any column $k \in \{1, \dots, K\}$, so $\mathbf{Q}$ must be non-negative.

Second, since we require $\tilde{\mathbf{B}} \geq 0$, and $\mathbf{B} \geq 0$, and $\tilde{\mathbf{B}} = \mathbf{Q}^{-1}\mathbf{B}$, and Equation (10) must hold, we know that $\mathbf{Q}^{-1} \geq 0$. The proof is analogous to the proof above.

Then, since $\mathbf{Q}$ and $\mathbf{Q}^{-1}$ are non-negative, Lemma 1.1 from Minc (1988) states that $\mathbf{Q}$ must be a generalized permutation matrix, i.e. there exists some diagonal matrix $\mathbf{D} \in \mathbb{R}^{K \times K}$ and permutation matrix $\mathbf{P} \in \mathbb{R}^{K \times K}$ such that $\mathbf{Q} = \mathbf{DP}$.

Third, since we require $\tilde{\mathbf{A}}\mathbf{1}_K = \mathbf{1}_N = \mathbf{A}\mathbf{Q}\mathbf{1}_K = \mathbf{1}_N$, we know that $\mathbf{Q}\mathbf{1}_K = \mathbf{1}_K$ and thus $\mathbf{Q} \in F(K, K)$ (i.e. $\mathbf{Q}$ must be a row-stochastic matrix)

Then, this means that $\mathbf{Q}$ must be a permutation matrix since

$$\begin{aligned} \mathbf{Q}\mathbf{1}_K &= \mathbf{1}_K \\ \rightarrow \mathbf{D}\mathbf{P}\mathbf{1}_K &= \mathbf{1}_K \\ \rightarrow \mathbf{D}\mathbf{1}_K &= \mathbf{1}_K \\ \rightarrow \mathbf{D} &= \mathbf{I}_K \end{aligned} \tag{111}$$

6.4 Proof for Convexity of Objective

If we measure the reconstruction error with the RSS, the objective in Eq. (4) is biconvex: it is convex in $\mathbf{A}$ when $\mathbf{B}$ is fixed and vice-versa. We prove convexity in $\mathbf{A}$; the argument for $\mathbf{B}$ is analogous.

Let $\mathbf{Z} := \mathbf{B}\mathbf{X} \in \mathbb{R}^{K \times D}$. With $\mathbf{B}$ fixed, Eq. (4) reduces to the quadratic programme

$$\hat{\mathbf{A}} = \arg \min_{\mathbf{A} \in \mathbb{R}^{N \times K}} \|\mathbf{X} - \mathbf{A}\mathbf{Z}\|_F^2 \quad \text{s.t.} \quad \mathbf{A} \geq 0, \mathbf{A}\mathbf{1}_K = \mathbf{1}_N. \tag{112}$$

As shown in Section 6.5, the feasible set is convex. We now rewrite the objective in canonical quadratic form.

Defining the vectorizations

$$\begin{aligned} \text{vec}(\mathbf{X}) &= [x_{11} \quad \dots \quad x_{1D} \quad \dots \quad x_{N1} \quad \dots \quad x_{ND}]^T \in \mathbb{R}^{ND} \\ \text{vec}(\mathbf{A}) &= [a_{11} \quad \dots \quad a_{1K} \quad \dots \quad a_{N1} \quad \dots \quad a_{NK}]^T \in \mathbb{R}_+^{NK} \end{aligned} \tag{113}$$

and Kronecker stack $\mathbf{Z}_\otimes$

$$\mathbf{Z}_\otimes = \begin{bmatrix} \mathbf{Z}^T & 0 & 0 & 0 \\ 0 & \mathbf{Z}^T & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \mathbf{Z}^T \end{bmatrix} = \mathbf{I}_N \otimes \mathbf{Z}^T \in \mathbb{R}^{ND \times NK} \tag{114}$$

we can rewrite the squared Frobenius norm as squared ℓ_2 -norm

$$\begin{aligned} \|\mathbf{X} - \mathbf{A}\mathbf{Z}\|_F^2 &= \|\text{vec}(\mathbf{X}) - \mathbf{Z}_\otimes \text{vec}(\mathbf{A})\|_2^2 \\ &= (\text{vec}(\mathbf{X}) - \mathbf{Z}_\otimes \text{vec}(\mathbf{A}))^T (\text{vec}(\mathbf{X}) - \mathbf{Z}_\otimes \text{vec}(\mathbf{A})) \\ &= \text{vec}(\mathbf{X})^T \text{vec}(\mathbf{X}) - 2 \text{vec}(\mathbf{X})^T \mathbf{Z}_\otimes \text{vec}(\mathbf{A}) + \text{vec}(\mathbf{A})^T \mathbf{Z}_\otimes^T \mathbf{Z}_\otimes \text{vec}(\mathbf{A}). \end{aligned} \tag{115}$$

Thus, the optimization problem can be written in the canonical form of a quadratic program

$$\arg \min_{\text{vec}(\mathbf{A}) \in \mathbb{R}^{NK}} \frac{1}{2} \text{vec}(\mathbf{A})^T \mathbf{Q} \text{vec}(\mathbf{A}) + \mathbf{c}^T \text{vec}(\mathbf{A}) \quad \text{s.t.} \quad \text{vec}(\mathbf{A}) \geq 0, \mathbf{C} \text{vec}(\mathbf{A}) = \mathbf{1}_N \tag{116}$$

with

$$\begin{aligned} \frac{1}{2} \mathbf{Q} &= \mathbf{Z}_\otimes^T \mathbf{Z}_\otimes = (\mathbf{I}_N \otimes \mathbf{Z}^T)^T (\mathbf{I}_N \otimes \mathbf{Z}^T) = (\mathbf{I}_N^T \otimes \mathbf{Z})(\mathbf{I}_N \otimes \mathbf{Z}^T) = \mathbf{I}_N^T \mathbf{I}_N \otimes \mathbf{Z}\mathbf{Z}^T = \mathbf{I}_N \otimes \mathbf{Z}\mathbf{Z}^T \\ \mathbf{c}^T &= -2 \text{vec}(\mathbf{X})^T \mathbf{Z}_\otimes \\ \mathbf{C} &= \mathbf{1}_K^T \otimes \mathbf{I}_N \end{aligned} \tag{117}$$

Because $\mathbf{Z}\mathbf{Z}^\top \succeq 0$, the Kronecker product $\mathbf{I}_N \otimes (\mathbf{Z}\mathbf{Z}^\top)$ is also positive semi-definite, and scaling by 2 preserves this property. Hence $\mathbf{Q} \succeq 0$.

The objective in (112) is a convex quadratic, and the constraints are linear; therefore the $\mathbf{A}$ -subproblem is a convex quadratic programme. Fixing $\mathbf{A}$ and repeating the same construction for $\mathbf{B}$ proves the corresponding convexity in $\mathbf{B}$.

6.5 Proof for Convexity of Standard Simplex

Let $\mathbf{x}, \mathbf{y} \in \Delta_{K-1}$, then for all $0 \leq \lambda \leq 1$, it is true that $\mathbf{z} = \lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in \Delta_{K-1}$

First we show that for all $k \in \{1, \dots, K\}$, $z_k \geq 0$. Recall that $z_k = \lambda x_k + (1 - \lambda)y_k$. Since $0 \leq \lambda \leq 1$, we have $1 - \lambda \geq 0$. Since $\mathbf{x}, \mathbf{y} \in \Delta_{K-1}$, we have $x_k \geq 0$ and $y_k \geq 0$. Thus $z_k = \lambda x_k + (1 - \lambda)y_k \geq 0$.

Then, we show that the elements of $\mathbf{z}$ sum to one.

$$\begin{aligned} \|\mathbf{z}\|_1 &= \sum_{k=1}^K z_k \\ &= \sum_{k=1}^K [\lambda x_k + (1 - \lambda)y_k] \\ &= \lambda \underbrace{\sum_{k=1}^K x_k}_{=1} + (1 - \lambda) \underbrace{\sum_{k=1}^K y_k}_{=1} \\ &= \lambda + 1 - \lambda \\ &= 1 \end{aligned}$$

6.6 Proof for Compactness of Standard Simplex

The standard $(K - 1)$ -simplex Δ_{K-1}

$$\Delta_{K-1} := \left\{ \mathbf{x} \in \mathbb{R}^K \mid \forall k \in \{1, \dots, K\}, x_k \geq 0 \wedge \sum_{k=1}^K x_k = 1 \right\}. \quad (118)$$

is a subset of a finite-dimensional Euclidean space. By the *Heine-Borel theorem*, a subset of $\mathbb{R}^K$ is compact if and only if it is both *closed* and *bounded*.

Step 1: Closedness.

We can write Δ_{K-1} as the finite intersection

$$\Delta_{K-1} = \left(\bigcap_{k=1}^K \underbrace{\{\mathbf{x} \in \mathbb{R}^K : x_k \geq 0\}}_{=: A_k} \right) \cap \underbrace{\{\mathbf{x} \in \mathbb{R}^K : \sum_{k=1}^K x_k = 1\}}_{=: B}. \quad (119)$$

- *Each A_k is closed.* Let $\pi_k : \mathbb{R}^K \rightarrow \mathbb{R}$, $\pi_k(\mathbf{x}) = x_k$. The coordinate projection π_k is linear, hence continuous. Because $[0, \infty) \subset \mathbb{R}$ is closed, its pre-image

$$A_k = \pi_k^{-1}([0, \infty)) \quad (120)$$

is closed. Geometrically each A_k is a closed half-space.

- B is closed. Define $g : \mathbb{R}^K \rightarrow \mathbb{R}$, $g(\mathbf{x}) = \sum_{k=1}^K x_k$. The set B can be written as $g^{-1}(\{1\})$. Since g is continuous and $\{1\}$ is closed in $\mathbb{R}$, B is closed.

Because a *finite* intersection of closed sets is closed, it follows that Δ_{K-1} is closed.

Step 2: Boundedness.

For $\mathbf{x} \in \Delta_{K-1}$ we have $0 \leq x_k \leq 1$ and $\sum_{k=1}^K x_k = 1$. Since $x_k^2 \leq x_k$ on $[0, 1]$,

$$\|\mathbf{x}\|_2^2 = \sum_{k=1}^K x_k^2 \leq \sum_{k=1}^K x_k = 1, \quad \implies \quad \|\mathbf{x}\|_2 \leq 1. \quad (121)$$

Hence every element of Δ_{K-1} lies in the closed Euclidean ball $B_2(0, 1)$. Hence, the simplex is bounded.

Step 3: Compactness.

Since Δ_{K-1} is both closed (Step 1) and bounded (Step 2), the Heine-Borel theorem implies that Δ_{K-1} is compact in $\mathbb{R}^K$.

6.7 Gradient of Vanilla Objective

To compute the gradient of the unconstrained objective with respect to $\mathbf{A}$ and $\mathbf{B}$, we first rewrite the residual sum of squares (Frobenius norm) in Equation (4) in terms of the trace

$$\begin{aligned} \text{RSS} &= \|\mathbf{X} - \mathbf{ABX}\|_F^2 \\ &= \text{tr} \left((\mathbf{X} - \mathbf{ABX})^T (\mathbf{X} - \mathbf{ABX}) \right) \\ &= \text{tr}(\mathbf{X}^T \mathbf{X}) - \text{tr}(\mathbf{X}^T \mathbf{ABX}) - \text{tr}(\mathbf{X}^T \mathbf{B}^T \mathbf{A}^T \mathbf{X}) + \text{tr}(\mathbf{X}^T \mathbf{B}^T \mathbf{A}^T \mathbf{ABX}) \\ &= \text{tr}(\mathbf{X}^T \mathbf{X}) - 2 \text{tr}(\mathbf{X}^T \mathbf{ABX}) + \text{tr}(\mathbf{X}^T \mathbf{B}^T \mathbf{A}^T \mathbf{ABX}) \end{aligned} \quad (122)$$

where we used that for any $\mathbf{G}, \mathbf{H} \in \mathbb{R}^{N \times N}$ it is true that $\text{tr}(\mathbf{G} + \mathbf{H}) = \text{tr}(\mathbf{G}) + \text{tr}(\mathbf{H})$ and $\text{tr}(\mathbf{G}^T) = \text{tr}(\mathbf{G})$

Next we will use Equation 101 from Petersen and Pedersen (2012) which states that for any matrices $G, H, J \in \mathbb{R}^{N \times N}$ we have

$$\frac{\partial}{\partial H} \text{tr}(GHJ) = G^T J^T \quad (123)$$

and Equation 116 which states that for any matrices $G, H, J \in \mathbb{R}^{N \times N}$ we have

$$\frac{\partial}{\partial H} \text{tr}(G^T H^T JHG) = J^T HGG^T + JHGG^T \quad (124)$$

So computing the gradient of the RSS with respect to $\mathbf{A}$ we have

$$\begin{aligned}
G^{(\mathbf{A})} &= \nabla_{\mathbf{A}} \text{RSS} \\
&= \nabla_{\mathbf{A}} [\text{tr}(\mathbf{X}^T \mathbf{X}) - 2 \text{tr}(\mathbf{X}^T \mathbf{A} \mathbf{B} \mathbf{X}) + \text{tr}(\mathbf{X}^T \mathbf{B}^T \mathbf{A}^T \mathbf{A} \mathbf{B} \mathbf{X})] \\
&= -2 \nabla_{\mathbf{A}} \text{tr}(\underbrace{\mathbf{X}^T}_G \underbrace{\mathbf{A}}_H \underbrace{\mathbf{B} \mathbf{X}}_J) + \nabla_{\mathbf{A}} \text{tr}(\underbrace{(\mathbf{B} \mathbf{X})^T}_{G^T} \underbrace{\mathbf{A}^T}_{H^T} \underbrace{\mathbf{I}}_J \underbrace{\mathbf{A}}_H \underbrace{\mathbf{B} \mathbf{X}}_G) \\
&= -2 \mathbf{X} \mathbf{X}^T \mathbf{B}^T + (\mathbf{I}^T \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T + \mathbf{I} \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T) \\
&= -2 \mathbf{X} \mathbf{X}^T \mathbf{B}^T + 2 \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T \\
&= 2 (\mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T - \mathbf{X} \mathbf{X}^T \mathbf{B}^T) \\
&= 2 (\mathbf{A} \mathbf{Z} \mathbf{Z}^T - \mathbf{X} \mathbf{Z}^T)
\end{aligned} \tag{125}$$

Similarly, computing the gradient of the RSS with respect to $\mathbf{B}$ we have

$$\begin{aligned}
G^{(\mathbf{B})} &= \nabla_{\mathbf{B}} \text{RSS} \\
&= \nabla_{\mathbf{B}} [\text{tr}(\mathbf{X}^T \mathbf{X}) - 2 \text{tr}(\mathbf{X}^T \mathbf{A} \mathbf{B} \mathbf{X}) + \text{tr}(\mathbf{X}^T \mathbf{B}^T \mathbf{A}^T \mathbf{A} \mathbf{B} \mathbf{X})] \\
&= -2 \nabla_{\mathbf{B}} \text{tr}(\underbrace{\mathbf{X}^T}_G \underbrace{\mathbf{A}}_H \underbrace{\mathbf{B} \mathbf{X}}_J) + \nabla_{\mathbf{B}} \text{tr}(\underbrace{\mathbf{X}^T}_{G^T} \underbrace{\mathbf{B}^T}_{H^T} \underbrace{\mathbf{A}^T \mathbf{A}}_J \underbrace{\mathbf{B}}_H \underbrace{\mathbf{X}}_G) \\
&= -2 \mathbf{A}^T \mathbf{X} \mathbf{X}^T + (\mathbf{A}^T \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T + \mathbf{A}^T \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T) \\
&= -2 \mathbf{A}^T \mathbf{X} \mathbf{X}^T + 2 \mathbf{A}^T \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T \\
&= 2 (\mathbf{A}^T \mathbf{A} \mathbf{B} \mathbf{X} \mathbf{X}^T - \mathbf{A}^T \mathbf{X} \mathbf{X}^T)
\end{aligned} \tag{126}$$

6.8 Gradient of l1-Normalization

Let $\mathbf{p} \in \mathbb{R}^K$. Define $\mathbf{r} \in \mathbb{R}^K$ element-wise as

$$r_k = \frac{\max(p_k, 0)}{\sum_{k''=1}^K \max(p_{k''}, 0)} = \frac{q_k}{\sum_{k''=1}^K q_{k''}} \tag{127}$$

where we define $q_k = \max(p_k, 0)$.

Then the gradient $\frac{d\mathbf{r}}{d\mathbf{p}} \in \mathbb{R}^{K \times K}$ is given by the chain rule

$$\frac{d\mathbf{r}}{d\mathbf{p}} = \frac{d\mathbf{r}}{d\mathbf{q}} \frac{d\mathbf{q}}{d\mathbf{p}} \tag{128}$$

The gradient of the rectified linear unit is given by

$$\frac{d\mathbf{q}}{d\mathbf{p}} = \text{diag} [\mathbb{1} [p_1 \geq 0] \quad \dots \quad \mathbb{1} [p_K \geq 0]]^T \tag{129}$$

While the gradient of the normalization is given by (see derivation below)

$$\frac{d\mathbf{r}}{d\mathbf{q}} = \frac{\left(\sum_{k''=1}^K q_{k''} \right) I_K - \mathbf{q} \mathbf{1}_K^T}{\left(\sum_{k''=1}^K q_{k''} \right)^2} \tag{130}$$

Thus we have

$$\frac{d\mathbf{r}}{d\mathbf{p}} = \frac{\left(\sum_{k''=1}^K q_{k''} \right) I_K - \mathbf{q} \mathbf{1}_K^T}{\left(\sum_{k''=1}^K q_{k''} \right)^2} \text{diag} [\mathbb{1} [p_1 > 0] \quad \dots \quad \mathbb{1} [p_K > 0]]^T \tag{131}$$

The gradient $\frac{d\mathbf{r}}{d\mathbf{p}} \in \mathbb{R}^{K \times K}$ has the form

$$\frac{d\mathbf{r}}{d\mathbf{q}} = \begin{bmatrix} \frac{\partial r_1}{\partial q_1} & \dots & \frac{\partial r_1}{\partial q_K} \\ \vdots & \ddots & \vdots \\ \frac{\partial r_K}{\partial q_1} & \dots & \frac{\partial r_K}{\partial q_K} \end{bmatrix} \quad (132)$$

The off-diagonal terms (i.e. $k \neq k'$) are given by

$$\begin{aligned} \frac{\partial r_k}{\partial q_{k'}} &= \frac{\partial}{\partial q_{k'}} \frac{q_k}{\sum_{k''=1}^K q_{k''}} \\ &= q_k \frac{\partial}{\partial q_{k'}} \left(\sum_{k''=1}^K q_{k''} \right)^{-1} \\ &= q_k (-1) \left(\sum_{k''=1}^K q_{k''} \right)^{-2} \underbrace{\frac{\partial}{\partial q_{k'}} \left(\sum_{k''=1}^K q_{k''} \right)}_{=1} \\ &= - \frac{q_k}{\left(\sum_{k''=1}^K q_{k''} \right)^2} \end{aligned} \quad (133)$$

To compute the diagonal terms we need to use the chain rule and can re-use the result from above

$$\begin{aligned} \frac{\partial r_k}{\partial q_k} &= \frac{\partial}{\partial q_k} \frac{q_k}{\sum_{k''=1}^K q_{k''}} \\ &= \frac{\partial}{\partial q_k} \left[\left(\frac{\partial}{\partial q_k} q_k \right) \left(\sum_{k''=1}^K q_{k''} \right)^{-1} \right] + \left[q_k \left(\frac{\partial}{\partial q_k} \left(\sum_{k''=1}^K q_{k''} \right)^{-1} \right) \right] \\ &= \left[1 \left(\sum_{k''=1}^K q_{k''} \right)^{-1} \right] + \left[- \frac{q_k}{\left(\sum_{k''=1}^K q_{k''} \right)^2} \right] \\ &= \frac{\sum_{k''=1}^K q_{k''}}{\left(\sum_{k''=1}^K q_{k''} \right)^2} - \frac{q_k}{\left(\sum_{k''=1}^K q_{k''} \right)^2} \end{aligned} \quad (134)$$

Now putting everything together we have

$$\frac{d\mathbf{r}}{d\mathbf{q}} = \frac{\left(\sum_{k''=1}^K q_{k''} \right) I_K - \mathbf{q} \mathbf{1}_K^T}{\left(\sum_{k''=1}^K q_{k''} \right)^2} \quad (135)$$

6.9 Gradient of Objective with Relaxed Archetype Constraints

Given the objective

$$\begin{aligned} \hat{\mathbf{A}}, \hat{\mathbf{B}} &= \arg \min_{\substack{\mathbf{A} \in \mathbb{R}^{N \times K} \\ \mathbf{B} \in \mathbb{R}^{K \times N}}} \|\mathbf{X} - \mathbf{A} \text{diag}(\boldsymbol{\alpha}) \mathbf{B} \mathbf{X}\|_F^2 \quad \text{subject to} \\ \mathbf{A} &\geq 0, \quad \mathbf{A} \mathbf{1}_K = \mathbf{1}_N \\ \mathbf{B} &\geq 0, \quad \mathbf{B} \mathbf{1}_N = \mathbf{1}_K \\ \forall k \in \{1, \dots, K\} \quad 1 - \delta &\leq \alpha_k \leq 1 + \delta \end{aligned} \quad (136)$$

we seek to derive the gradient with respect to $\mathbf{A}$, $\mathbf{B}$, and α . Following section 6.7 we first rewrite the objective in terms of the trace.

$$\begin{aligned}
\text{RSS} &= \|\mathbf{X} - \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}\|_F^2 \\
&= \text{tr} \left((\mathbf{X} - \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X})^T (\mathbf{X} - \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}) \right) \\
&= \text{tr} (\mathbf{X}^T \mathbf{X}) - \text{tr} (\mathbf{X}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}) - \text{tr} (\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \mathbf{A}^T \mathbf{X}) \\
&\quad + \text{tr} (\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}) \\
&= \text{tr} (\mathbf{X}^T \mathbf{X}) - 2 \text{tr} (\mathbf{X}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}) + \text{tr} (\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X})
\end{aligned} \tag{137}$$

We will use again the identities from Equation 123 and Equation 124

Computing the gradient of the RSS with respect to $\mathbf{A}$ we have

$$\begin{aligned}
G^{(A)} &= \nabla_{\mathbf{A}} \text{RSS} \\
&= \nabla_{\mathbf{A}} [\text{tr} (\mathbf{X}^T \mathbf{X}) - 2 \text{tr} (\mathbf{X}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}) + \text{tr} (\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X})] \\
&= -2 \nabla_{\mathbf{A}} \text{tr} (\underbrace{\mathbf{X}^T}_G \underbrace{\mathbf{A}}_H \underbrace{\text{diag}(\alpha) \mathbf{B} \mathbf{X}}_J) + \nabla_{\mathbf{A}} \text{tr} (\underbrace{(\text{diag}(\alpha) \mathbf{B} \mathbf{X})^T}_{G^T} \underbrace{\mathbf{A}^T}_{H^T} \underbrace{\mathbf{I}}_J \underbrace{\mathbf{A}}_H \underbrace{\text{diag}(\alpha) \mathbf{B} \mathbf{X}}_G) \\
&= -2 \mathbf{X} \mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) + 2 \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \\
&= 2 (\mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) - \mathbf{X} \mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha))
\end{aligned} \tag{138}$$

Computing the gradient of the RSS with respect to $\mathbf{B}$ we have

$$\begin{aligned}
G^{(B)} &= \nabla_{\mathbf{B}} \text{RSS} \\
&= \nabla_{\mathbf{B}} [\text{tr} (\mathbf{X}^T \mathbf{X}) - 2 \text{tr} (\mathbf{X}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}) + \text{tr} (\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X})] \\
&= -2 \nabla_{\mathbf{B}} \text{tr} (\underbrace{\mathbf{X}^T \mathbf{A} \text{diag}(\alpha)}_G \underbrace{\mathbf{B}}_H \underbrace{\mathbf{X}}_J) + \nabla_{\mathbf{B}} \text{tr} (\underbrace{\mathbf{X}^T}_G \underbrace{\mathbf{B}^T}_{H^T} \underbrace{\text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha)}_J \underbrace{\mathbf{B}}_H \underbrace{\mathbf{X}}_G) \\
&= -2 \text{diag}(\alpha) \mathbf{A}^T \mathbf{X} \mathbf{X}^T + 2 \text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X} \mathbf{X}^T \\
&= 2 [\text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X} \mathbf{X}^T - \text{diag}(\alpha) \mathbf{A}^T \mathbf{X} \mathbf{X}^T]
\end{aligned} \tag{139}$$

Finally, computing the gradient of the RSS with respect to α we have

$$\begin{aligned}
G^{(\alpha)} &= \nabla_{\alpha} \text{RSS} \\
&= \nabla_{\alpha} [\text{tr} (\mathbf{X}^T \mathbf{X}) - 2 \text{tr} (\mathbf{X}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X}) + \text{tr} (\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X})] \\
&= -2 \nabla_{\alpha} \text{tr} (\underbrace{\mathbf{X}^T \mathbf{A} \text{diag}(\alpha)}_G \underbrace{\mathbf{B} \mathbf{X}}_H) + \nabla_{\alpha} \text{tr} (\underbrace{\mathbf{X}^T \mathbf{B}^T}_{G^T} \underbrace{\text{diag}(\alpha)}_{H^T} \underbrace{\mathbf{A}^T \mathbf{A} \text{diag}(\alpha)}_J \underbrace{\mathbf{B} \mathbf{X}}_H) \\
&= -2 \mathbf{A}^T \mathbf{X} \mathbf{X}^T \mathbf{B}^T + 2 \mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T \\
&= 2 [\mathbf{A}^T \mathbf{A} \text{diag}(\alpha) \mathbf{B} \mathbf{X} \mathbf{X}^T \mathbf{B}^T - \mathbf{A}^T \mathbf{X} \mathbf{X}^T \mathbf{B}^T]
\end{aligned} \tag{140}$$

6.10 Gradient of Objective with Relaxed Archetype Constraints and Core-sets

Given the objective

$$\begin{aligned}
\hat{\mathbf{A}}, \hat{\mathbf{B}} &= \arg \min \|\mathbf{W} \tilde{\mathbf{X}} - \mathbf{W} \mathbf{A} \text{diag}(\alpha) \mathbf{B} \tilde{\mathbf{X}}\|_F^2 \quad \text{subject to} \\
\mathbf{A} &\in F(\tilde{N}, K) \\
\mathbf{B} &\in F(K, \tilde{N}) \\
\forall k \in \{1, \dots, K\} \quad 1 - \delta &\leq \alpha_k \leq 1 + \delta
\end{aligned} \tag{141}$$

we seek to derive the gradient with respect to $\mathbf{A}$, $\mathbf{B}$, and α .

To simplify the derivation we define $\check{\mathbf{A}} = \mathbf{W}\mathbf{A}$ and $\check{\mathbf{X}} = \mathbf{W}\mathbf{X}$. Following section 6.7 we first rewrite the objective in terms of the trace.

$$\begin{aligned}
\text{RSS} &= \|\check{\mathbf{X}} - \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}}\|_F^2 \\
&= \text{tr} \left(\left(\check{\mathbf{X}} - \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right)^T \left(\check{\mathbf{X}} - \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right) \right) \\
&= \text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{X}} \right) - \text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right) - \text{tr} \left(\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{X}} \right) \\
&\quad + \text{tr} \left(\mathbf{X}^T \mathbf{B}^T \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \mathbf{X} \right) \\
&= \text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{X}} \right) - 2 \text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right) + \text{tr} \left(\check{\mathbf{X}}^T \mathbf{B}^T \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right)
\end{aligned} \tag{142}$$

Using the identities from Equation 123 and Equation 124, the gradient of the RSS with respect to $\mathbf{B}$ is given by

$$\begin{aligned}
G^{(B)} &= \nabla_{\mathbf{B}} \text{RSS} \\
&= \nabla_{\mathbf{B}} \left[\text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{X}} \right) - 2 \text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right) + \text{tr} \left(\check{\mathbf{X}}^T \mathbf{B}^T \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right) \right] \\
&= -2 \nabla_{\mathbf{B}} \text{tr} \left(\underbrace{\check{\mathbf{X}}^T \check{\mathbf{A}} \text{diag}(\alpha)}_G \underbrace{\mathbf{B}}_H \underbrace{\check{\mathbf{X}}}_J \right) + \nabla_{\mathbf{B}} \text{tr} \left(\underbrace{\check{\mathbf{X}}^T}_{G^T} \underbrace{\mathbf{B}^T}_{H^T} \underbrace{\text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha)}_J \underbrace{\mathbf{B}}_H \underbrace{\check{\mathbf{X}}}_G \right) \\
&= -2 \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{X}} \check{\mathbf{X}}^T + 2 \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \check{\mathbf{X}}^T \\
&= 2 \left[\text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \check{\mathbf{X}}^T - \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{X}} \check{\mathbf{X}}^T \right]
\end{aligned} \tag{143}$$

Computing the gradient of the RSS with respect to α we have

$$\begin{aligned}
G^{(\alpha)} &= \nabla_{\alpha} \text{RSS} \\
&= \nabla_{\alpha} \left[\text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{X}} \right) - 2 \text{tr} \left(\check{\mathbf{X}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right) + \text{tr} \left(\check{\mathbf{X}}^T \mathbf{B}^T \text{diag}(\alpha) \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \right) \right] \\
&= -2 \nabla_{\alpha} \text{tr} \left(\underbrace{\check{\mathbf{X}}^T \check{\mathbf{A}} \text{diag}(\alpha)}_G \underbrace{\mathbf{B} \check{\mathbf{X}}}_H \right) + \nabla_{\alpha} \text{tr} \left(\underbrace{\check{\mathbf{X}}^T \mathbf{B}^T}_{G^T} \underbrace{\text{diag}(\alpha)}_{H^T} \underbrace{\check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha)}_J \underbrace{\mathbf{B} \check{\mathbf{X}}}_H \right) \\
&= -2 \check{\mathbf{A}}^T \check{\mathbf{X}} \check{\mathbf{X}}^T \mathbf{B}^T + 2 \check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \check{\mathbf{X}}^T \mathbf{B}^T \\
&= 2 \left[\check{\mathbf{A}}^T \check{\mathbf{A}} \text{diag}(\alpha) \mathbf{B} \check{\mathbf{X}} \check{\mathbf{X}}^T \mathbf{B}^T - \check{\mathbf{A}}^T \check{\mathbf{X}} \check{\mathbf{X}}^T \mathbf{B}^T \right]
\end{aligned} \tag{144}$$